

First Pattern Failure in Catalan Permutations

A proof of OEIS A273821, exact formulas,
all-orders expansions, and a critical phase transition

Research report prepared for Vladimir Reshetnikov
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Abstract

For a uniformly chosen 123-avoiding permutation of $[n]$, let K_n be the largest number of its largest entries whose induced permutation still avoids 132. The generating function for the joint counts $T(n, k)$ is marked conjectural in OEIS A273821. We prove it by inserting successive minima, then telescope the resulting Catalan convolution to obtain a difference of two binomial coefficients. This resolves both the posted generating-function conjecture and the indicated growth-rate question for every nonzero fixed column.

The exact formula yields substantially more: every column $k \geq 2$ is hypergeometric in n ; its full inverse-power expansion is computable to any order; and K_n has an explicit discrete limit law with exponentially weighted total-variation error $O(n^{-1})$. The fixed-column approximation changes at $k \asymp \sqrt{n}$, where we prove an explicit crossover. For k proportional to n , two distinct exponential sectors are separated exactly rather than hidden under an algebraic error term.

Weighting permutations by y^{K_n} produces a transition at $y = 2$. Below the transition K_n stays bounded; above it $n - K_n$ stays bounded; at the transition K_n/n converges to $\text{Beta}(1, \frac{1}{2})$. We give an exact critical partition function and mean, a critical window, Lambert- W inverse expansions, independent computation, and a formalization and research agenda.

Status. The report supplies conventional proofs and executable checks, not a Lean formalization. The identified OEIS conjecture is proved here; priority for every extension is not certified by an exhaustive literature search. No OEIS edit or repository change has been submitted.

Editorial note (5 October 2026, batch 98). Since batch 98 the report has two Parts. Part I is the report described on the title page: manuscript 04 of batch 74, on the pattern 132. Part II (Section 14 onwards) adds manuscript 07 of batch 98, which answers Part I's first research question: it treats the patterns $1(r+1)r \cdots 2$ for every $r \geq 2$, with exact enumeration for all r and, for every fixed $r \geq 3$, a critical law of a different kind (a tight, heavy-tailed deficit instead of the beta law), a displaced coexistence window and local complex zeros. Its labels carry the prefix `fpf:lp:`. Appendices A and B, after Part II, belong to Part I.

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Part I

The pattern 132: proof of the A273821 conjecture and a critical phase transition

1 The question, the answer, and the scope of the claim

1.1 The statistic and the live OEIS question

A permutation avoids a pattern if no subsequence, with positions increasing, has the same relative order as that pattern. Thus 123 avoidance forbids three increasing entries, and 132 avoidance forbids indices $i < j < h$ with $\pi_i < \pi_h < \pi_j$. Write $\text{Av}_n(123)$ for the first class.

For $\pi \in \text{Av}_n(123)$, restrict π to its k largest values without changing their order. Let $K(\pi)$ be the largest k for which this restriction avoids 132. Set

$$T(n, k) = \#\{\pi \in \text{Av}_n(123) : K(\pi) = k\}, \quad Z_n(y) = \sum_{k=1}^n T(n, k)y^k. \quad (1.1)$$

The triangle is A273821, introduced by David Callan in 2016 [1]. As retrieved on October 1, 2026, its bivariate generating function remains labeled conjectural; its comments also suggest growth constant 4 for each column other than the first. These are the two specific posted questions answered below.

We use $C_n = \frac{1}{n+1} \binom{2n}{n}$ and

$$C(x) = \sum_{n \geq 0} C_n x^n = \frac{2}{1 + \sqrt{1 - 4x}}, \quad C(x) = 1 + xC(x)^2. \quad (1.2)$$

Our insertion argument will itself establish the Catalan row sums, so the proof does not assume the conjectural triangle recurrence or generating function. All entries outside $1 \leq k \leq n$ are zero. For a binomial coefficient with nonnegative upper argument, a negative lower argument is interpreted as zero.

1.2 Principal conclusions

The central enumeration result is

$$T(n, k) = 2^{k-1} \binom{2n-2k}{n-k-1} - \binom{2n-k-1}{n-k-1} \quad (n > k \geq 1), \quad (1.3)$$

with the separate diagonal value $T(n, n) = 2^{n-1}$. In particular, the first column is zero after its initial entry.

The resulting bivariate generating function is

$$F(x, y) := \sum_{n \geq 1} Z_n(y)x^n = \frac{xy}{1-2xy} + \frac{x^3 y^2 C(x)^3}{(1-2xy)(1-xyC(x))}. \quad (1.4)$$

This is algebraically the conjectured OEIS expression. Its present form records the first failure of 132 avoidance explicitly.

For every fixed $k \geq 2$,

$$T(n, k) \sim \frac{(k-1)(k+4)}{2^{k+3}\sqrt{\pi}} \frac{4^n}{n^{3/2}}. \quad (1.5)$$

We prove a full expansion with explicit first coefficients and an all-orders recursion, not only this leading equivalent. Moreover, for a uniform permutation in $\text{Av}_n(123)$,

$$\mathbb{P}(K_n = k) \longrightarrow p_k := \frac{(k-1)(k+4)}{2^{k+3}} \quad (k \geq 2), \quad \sum_{k \geq 2} p_k y^k = \frac{y^2(3-y)}{2(2-y)^3}. \quad (1.6)$$

There is a second, macroscopic phenomenon. Under the probability measure proportional to $y^{K(\pi)}$, the scaling of K_n changes at $y = 2$:

$$\begin{array}{l|l} 0 < y < 2 & K_n \text{ converges to an explicit discrete law,} \\ y = 2 & K_n/n \Rightarrow \text{Beta}(1, \frac{1}{2}), \\ y > 2 & n - K_n \text{ converges to an explicit discrete law.} \end{array} \quad (1.7)$$

The critical normalization has the exact evaluation

$$Z_n(2) = (n+2) \binom{2n}{n} - 4^n. \quad (1.8)$$

These conclusions are consequences of the same exact first-failure formula, not independent guesses from numerical plots.

1.3 What is established, and what is not claimed

The contribution is a self-contained resolution of an identifiable posted conjecture, together with a detailed analysis of its refined counting triangle. Pattern-avoiding permutations, Catalan generating trees, Stirling expansions, and singularity analysis are established methods; they are not claimed as new here. The two low columns already have known companion entries, [A000245](#) and [A071718](#), including known leading asymptotics [2, 3].

A targeted search of the sequence identifier and defining statistic did not locate an independent published proof of the A273821 conjecture. That is a limited search result, not a proof of priority. Accordingly, “proved here” is used literally, while an assertion that every extension is unprecedented is deliberately avoided.

The ProveIt repository has an independent transseries library for asymptotic scales and differential blocks, together with an inversion volume, a combinatorial companion, and verification programs [9]. Here we distinguish formal expansions, analytic interpretation, and executable checks, especially when an exponentially smaller sector is present. No theorem from the repository is needed as an unverified premise, and none of the results below is represented as already formalized in Lean.

1.4 Provenance and place in the collection

[write] Added 1 October 2026, batch 74. This report is a single manuscript, printed in full: manuscript 04 of batch 74 of ProveIt’s incoming-reports intake, delivered as the archive `A273821_First_Pattern_Failure.zip` in commit `c664fc2f0` and placed in the research-report collection (`SetTheory/Cardinals/docs/reports/`

enumerative-combinatorics/) in commit b669cff87. Its pin is commit 291f6e6bb, at which its author read the root README and, from the default branch, Analysis/Transseries/README.md (blob 3d9ce016). The writing step changed only the labels (every label now carries the prefix `fpf:`), added these [write] notes and six section labels, and added two repository references [7, 8]; no statement or proof was altered.

No other report in the collection treats A273821 or its statistic. The nearest relatives are of the same genre only: `enumerative-combinatorics/mesh-avoidance-catalan-inflation` proves a Catalan-type generating function marked conjectural in OEIS A289587, and `enumerative-combinatorics/adjacency-bounded-132-avoiders` studies 132-avoiders under a different constraint. Neither shares a theorem with this report.

The transseries material mentioned above remains methodological context only. The inverse expansion of Section 11 is an instance of the balanced inverse theorem of the combinatorial-inverses volume [7], and the rounding discussion is an instance of the staircase theorem of the transseries-and-inversion volume [8]; neither is claimed as new. Placement in the collection, beside the repository's Lean and Rocq developments, confers no formal status: nothing in this report is formalized.

[write] **Added 5 October 2026, batch 98.** The note above describes Part I, which is still printed in full. Since batch 98 the report also has Part II, built from a second manuscript; its provenance is in Section 14.1. The statement that no other report of the collection treats A273821 or its statistic remains true.

2 A minimum-insertion model for Catalan permutations

2.1 The active suffix

Every nonempty permutation has a unique parent obtained by deleting its minimum and standardizing the remaining values. Reversing this operation builds a permutation by inserting successively smaller values.

Definition 2.1. The suffix state d of a nonempty permutation is the length of its longest terminal decreasing subsequence of *consecutive positions*. Thus the last d entries decrease, and either the permutation is decreasing or the entry immediately before them forms an ascent with the first of those d entries. An insertion site is indexed by the number j of old entries to its right.

Lemma 2.2 (Active sites). *For a permutation in $\text{Av}_n(123)$ with suffix state d , insertion of a new minimum preserves 123 avoidance exactly at the sites $0 \leq j \leq d$. The new state is $d + 1$ when $j = 0$, and is j when $1 \leq j \leq d$. Consequently the state production rule is*

$$d \longrightarrow 1, 2, \dots, d, d + 1, \quad (2.1)$$

with each child state occurring once.

Proof. Any newly formed increasing triple must start at the new minimum. Such a triple exists precisely when the entries to its right contain an increasing pair. Avoiding such a pair means that this right-hand suffix is decreasing, which is equivalent to $j \leq d$. If the minimum is appended, it lengthens the decreasing suffix by one. Otherwise it lies just before a decreasing suffix of length j and is smaller than its first entry, so that suffix cannot extend to the left. Its new length is exactly j . $\square$

2.2 Descendant generating functions

Let $D_d(x)$ count descendants of a state- d permutation, with x marking the number of additional minima inserted. The particular values or earlier shape of the permutation do not matter, because the production rule depends only on d .

Lemma 2.3. *For every $d \geq 1$,*

$$D_d(x) = C(x)^{d+1}. \quad (2.2)$$

Proof. The production rule gives

$$D_d(x) = 1 + x \sum_{j=1}^{d+1} D_j(x).$$

This infinite system has a unique formal-power-series solution with constant term 1: after the coefficients through degree $m - 1$ have been determined for every state, the equation determines the coefficient of degree m for every state using only a finite sum. To verify the claimed solution, use $xC^2 = C - 1$:

$$x \sum_{j=1}^{d+1} C^{j+1} = (C - 1)(1 + C + \cdots + C^d) = C^{d+1} - 1.$$

Uniqueness finishes the proof. $\square$

The single-entry permutation has state 1. Therefore the number of size- n permutations generated is

$$[x^{n-1}]D_1(x) = [x^{n-1}]C(x)^2 = C_n,$$

where the final equality follows from $xC^2 = C - 1$. Thus this construction also proves the Catalan row sums.

2.3 The two safe sites for avoiding both patterns

Suppose the current permutation avoids both 123 and 132. At an active site the entries to the right of a newly inserted minimum decrease. The new minimum starts a 132 occurrence exactly when there are at least two such entries. Hence the insertions that remain safe for both patterns are precisely $j = 0$ and $j = 1$.

The first appends the new minimum and increases d by one; the second places it before the last entry and resets d to 1. These two choices encode all size- k safe prefixes by binary words of length $k - 1$.

Lemma 2.4. *Let $a_{k,d}$ be the number of permutations of size k avoiding both patterns and having suffix state d . Then*

$$a_{k,k} = 1, \quad a_{k,d} = 2^{k-d-1} \quad (1 \leq d < k). \quad (2.3)$$

In particular, there are 2^{k-1} such permutations, and for $2 \leq j \leq k$,

$$\sum_{d=j}^k a_{k,d} = 2^{k-j}. \quad (2.4)$$

Proof. The state is k only if every insertion appends. Otherwise locate the last reset: to finish at state $d < k$ it must be followed by exactly $d - 1$ appends. The preceding $k - d - 1$ decisions are free. Summation gives both remaining assertions. $\square$

3 First failure and the conjectured generating function

3.1 The first-failure decomposition

Once a 132 occurrence has appeared among the largest entries, inserting smaller entries cannot remove it. Thus $K = k < n$ is equivalent to remaining safe through size k and failing at the next insertion. At that failure insertion the active site has $j \geq 2$ entries to its right.

Theorem 3.1 (Positive Catalan convolution). *For $n > k \geq 2$,*

$$T(n, k) = \sum_{j=2}^k 2^{k-j} [x^{n-k-1}] C(x)^{j+1}. \quad (3.1)$$

For $n = k$, $T(n, n) = 2^{n-1}$; for $n > 1$, $T(n, 1) = 0$.

Proof. For a fixed failing site j , a safe size- k prefix permits that insertion exactly when its state d satisfies $d \geq j$. There are 2^{k-j} such prefixes by Lemma 2.4. The failing insertion gives state j by Lemma 2.2; the remaining $n - k - 1$ insertions are arbitrary 123-preserving descendants and are counted by C^{j+1} . The parent operation is unique, so these cases are disjoint and exhaustive. The boundary cases follow from the safe-prefix count and the fact that a 132 occurrence needs three entries. $\square$

For example, the row at size 5 is

$$(T(5, 1), \dots, T(5, 5)) = (0, 9, 10, 7, 16).$$

The off-diagonal formula gives $T(5, 3) = [x](2C^3 + C^4) = 6 + 4 = 10$. The row sums to $42 = C_5$. This example is computed from the insertion argument, not used to infer it.

[write] Added 5 October 2026, batch 98. Part II proves the analogue for the patterns $1(r+1)r \cdots 2$ by the same insertion argument: Theorem 16.1, with the positive formula (16.4), which is (3.1) at $r = 2$ (Table 4).

3.2 Summation of the columns

Theorem 3.2 (Proof of the posted generating-function conjecture). *The identity (1.4) holds in $\mathbb{Q}[y][[x]]$. It is equal to*

$$C(x) - 1 + \frac{(1-y)(1-xy)\{1 - (1-xy)C(x)\}}{(1-2xy)(1-y+xy^2)}, \quad (3.2)$$

the expression marked conjectural in OEIS A273821 [1].

Proof. The size generating function for column k is

$$G_k(x) = 2^{k-1}x^k + x^{k+1} \sum_{j=2}^k 2^{k-j} C(x)^{j+1}, \quad G_1(x) = x. \quad (3.3)$$

The diagonal contribution sums to $xy/(1-2xy)$. In the second contribution, write $k = j + h$ and sum the two geometric series:

$$\begin{aligned} \sum_{k \geq 2} y^k x^{k+1} \sum_{j=2}^k 2^{k-j} C^{j+1} &= xC \sum_{j \geq 2} (xyC)^j \sum_{h \geq 0} (2xy)^h \\ &= \frac{x^3 y^2 C^3}{(1-2xy)(1-xyC)}. \end{aligned}$$

This proves (1.4). To compare with (3.2), replace xC^2 by $C - 1$ and use

$$1 - y + xy^2 = (1 - xyC)(1 - y/C).$$

Clearing denominators then gives identical numerators. This is an identity of rational algebraic functions; any removable factors in the displayed OEIS form are interpreted through the formal series (1.4). $\square$

Setting $y = 1$ gives $F(x, 1) = C(x) - 1$, consistently with the combinatorial row-sum proof. The derivation also shows why two apparently different singular factors occur: $1 - 2xy$ records the binary safe-prefix evolution, whereas $1 - xyC$ records the active suffix at first failure.

4 Two binomials, exact tails, and hypergeometric columns

4.1 A telescoping Catalan convolution

Lagrange inversion applied to $C = 1 + xC^2$ gives, for $m \geq 0$ and $j \geq 1$,

$$b(m, j) := [x^m]C(x)^{j+1} = \frac{j+1}{2m+j+1} \binom{2m+j+1}{m} = \binom{2m+j}{m} - \binom{2m+j}{m-1}. \quad (4.1)$$

For completeness, if $m > 0$, set $u = C - 1 = x(1 + u)^2$. The Lagrange formula for $[x^m](1 + u)^{j+1}$ is $\frac{j+1}{m} [u^{m-1}](1 + u)^{2m+j}$, which simplifies to the first expression. At $m = 0$ both sides are 1.

Lemma 4.1 (Telescoper). *For $t_j = 2^{-j} \binom{2m+j}{m}$,*

$$2^{-j} b(m, j) = 2(t_j - t_{j+1}). \quad (4.2)$$

Proof. Pascal's identity gives

$$2 \binom{2m+j}{m} - \binom{2m+j+1}{m} = \binom{2m+j}{m} - \binom{2m+j}{m-1}.$$

Multiplication by 2^{-j} is exactly the claim. $\square$

Theorem 4.2 (Exact entries and cumulative counts). *The binomial formula (1.3) holds. In addition, for $1 \leq k \leq n$,*

$$S(n, k) := \sum_{h=k}^n T(n, h) = 2^{k-1} \left(\binom{2n-2k+1}{n-k} - \binom{2n-k}{n-k-1} \right). \quad (4.3)$$

Proof. In (3.1) put $m = n - k - 1$. Lemma 4.1 telescopes the sum to $2^{k+1}(t_2 - t_{k+1})$, giving (1.3). For $k = 1 < n$, its two terms coincide, as required.

For the cumulative count, being safe through size k is the only restriction. Put $\ell = n - k$. Lemmas 2.3 and 2.4 give

$$S(n, k) = [x^\ell] \left(C^{k+1} + \sum_{d=1}^{k-1} 2^{k-d-1} C^{d+1} \right).$$

With $m = \ell$, the telescoping sum over d is $2^k(t_1 - t_k)$, and the remaining term is $b(\ell, k)$. The result is $2^{k-1} \binom{2\ell+1}{\ell} - \binom{2\ell+k}{\ell-1}$, which is (4.3). It includes $k = n$ under the stated binomial convention. $\square$

In particular $S(n, 1) = C_n$. The positive formula (3.1) proves positivity of the difference of binomials for $2 \leq k < n$, without needing a cancellation estimate.

4.2 A polynomial factor and a first-order recurrence

Definition 4.3. For $k \geq 2$, define

$$H_k(n) = 2^{k-1}(n)_{\underline{k-1}} - (2n - k - 1)_{\underline{k-1}}, \quad (a)_{\underline{r}} = \prod_{j=0}^{r-1} (a - j). \quad (4.4)$$

Theorem 4.4. The polynomial H_k has degree $k - 2$, with leading coefficient $2^{k-3}(k - 1)(k + 4)$. For $n > k$,

$$T(n, k) = \frac{(2n - 2k)!}{(n - k - 1)! n!} H_k(n). \quad (4.5)$$

Consequently each fixed off-diagonal column is hypergeometric in n , with recurrence

$$\begin{aligned} (n - k)(n + 1)H_k(n)T(n + 1, k) \\ = 2(n - k + 1)(2n - 2k + 1)H_k(n + 1)T(n, k), \quad n > k. \end{aligned} \quad (4.6)$$

Proof. Factor the factorial expression in (1.3) to obtain (4.5). Both terms of H_k have leading term $2^{k-1}n^{k-1}$, so those terms cancel. Expanding the next coefficient by summing the roots of the two products gives $2^{k-3}(k - 1)(k + 4) > 0$. Thus the degree is exactly $k - 2$. Taking the ratio of the factorial prefactors at $n + 1$ and n gives (4.6). Positivity follows from Theorem 3.1 and the positive prefactor. $\square$

The first examples are

$$H_2(n) = 3, \quad H_3(n) = 14n - 20, \quad H_4(n) = 48n^2 - 198n + 210. \quad (4.7)$$

This first-order structure is stronger than merely knowing that each column is D-finite.

4.3 Known companion sequences and their offsets

The identifications noted in the OEIS entry become exact with the following ranges:

$$\begin{aligned} T(n, 2) &= \mathbf{A0000245}(n - 2), & n &\geq 3, \\ T(n, 3) &= \mathbf{A071718}(n - 3), & n &\geq 4. \end{aligned} \quad (4.8)$$

For the second column, (3.3) gives $G_2 = 2x^2 + x^3C^3$; the companion generating function is xC^3 . For the third, $G_3 = 4x^3 + x^4(2C^3 + C^4)$, and

$$1 + x(2C^3 + C^4) = C^3 + x^2C^4 = (1 + x^2C)C^3,$$

which is the companion generating function. These identities follow using $xC^2 = C - 1$. The initial diagonal exceptions are important: $T(2, 2) = 2$ instead of the companion's initial 0, and $T(3, 3) = 4$ instead of its initial 1. The two companion leading equivalents already appear in their entries and are not novelty claims of this report [2, 3].

5 All-orders fixed-column asymptotics

5.1 A rational ratio to the Catalan numbers

Proposition 5.1. *For each $k \geq 2$ and $n > k$, the ratio $T(n, k)/C_n$ is rational in n :*

$$\begin{aligned} \frac{T(n, k)}{C_n} &= (n+1) \frac{2^{k-1} (n)_{\underline{k+1}} (n)_{\underline{k-1}}}{(2n)_{\underline{2k}}} \\ &\quad - (n+1) \frac{(n)_{\underline{k+1}}}{(2n)_{\underline{k+1}}}. \end{aligned} \quad (5.1)$$

Proof. Divide each binomial term of (1.3) by $(2n)!/(n!^2(n+1))$ and cancel the factorials. $\square$

The two terms on the right individually grow linearly with n for fixed k , but their difference has a finite limit. Resolving that cancellation before expanding is useful both algebraically and numerically.

5.2 An explicit coefficient engine

Put $t = 1/n$ and $S_m(q) = \sum_{r=0}^q r^m$. For $m \geq 1$ define

$$\begin{aligned} a_m(k) &= \frac{2^{-m} S_m(2k-1) - S_m(k) - S_m(k-2)}{m}, \\ b_m(k) &= \frac{(2^{-m} - 1) S_m(k)}{m}, \end{aligned} \quad (5.2)$$

and let $A_k(t) = \sum_{m \geq 1} a_m(k) t^m$, $B_k(t) = \sum_{m \geq 1} b_m(k) t^m$. Taking logarithms of the finite products in (5.1) yields the local identity

$$\frac{T(n, k)}{C_n} = 2^{-k-1} \frac{1+t}{t} (e^{A_k(t)} - e^{B_k(t)}). \quad (5.3)$$

For fixed k , this is the Taylor expansion of a rational function at $t = 0$ after a removable singularity is canceled. The series therefore converges for sufficiently small $|t|$, unlike the separate Stirling series used for C_n .

The leading difference is

$$d_1(k) := a_1(k) - b_1(k) = \frac{(k-1)(k+4)}{4}. \quad (5.4)$$

Exponentials of either logarithmic series can be generated without repeated symbolic exponentiation: if $E(t) = \exp(\sum_{m \geq 1} \alpha_m t^m) = \sum E_j t^j$, then

$$E_0 = 1, \quad E_j = \frac{1}{j} \sum_{m=1}^j m \alpha_m E_{j-m}. \quad (5.5)$$

This supplies every probability correction as a rational function of k .

To include the Catalan prefactor, write the formal Stirling logarithm

$$L_C(t) = -\log(1+t) + \sum_{r \geq 1} \frac{B_{2r}(2^{1-2r} - 2)}{2r(2r-1)} t^{2r-1}, \quad (5.6)$$

where B_{2r} are Bernoulli numbers with $B_2 = 1/6$. Stirling's expansion [5] gives

$$C_n \sim \frac{4^n}{\sqrt{\pi n^{3/2}}} e^{L_C(1/n)} = \frac{4^n}{\sqrt{\pi n^{3/2}}} \left(1 - \frac{9}{8n} + \frac{145}{128n^2} - \frac{1155}{1024n^3} + \cdots \right). \quad (5.7)$$

The symbol $\sim$ here means agreement at every fixed truncation order; it does not assert convergence of the infinite Stirling series.

Theorem 5.2 (Full fixed-column expansion). *For every fixed integer $k \geq 2$ and every $J \geq 0$,*

$$T(n, k) = \frac{p_k}{\sqrt{\pi}} \frac{4^n}{n^{3/2}} \left(\sum_{j=0}^J \frac{c_j(k)}{n^j} + O_{k,J}(n^{-J-1}) \right), \quad p_k = \frac{(k-1)(k+4)}{2^{k+3}}, \quad (5.8)$$

where $c_0 = 1$ and the formal generating series of the coefficients is

$$\sum_{j \geq 0} c_j(k) t^j = e^{L_C(t)} \frac{(1+t)(e^{A_k(t)} - e^{B_k(t)})}{d_1(k)t}. \quad (5.9)$$

In particular,

$$\begin{aligned} c_1(k) &= -\frac{k^3 - k^2 - 17k + 36}{8(k+4)}, \\ c_2(k) &= \frac{4k^5 - 32k^4 - 22k^3 + 578k^2 - 2601k + 1740}{384(k+4)}. \end{aligned} \quad (5.10)$$

Proof. Equations (5.3) and (5.4) give the convergent fixed- k ratio expansion with leading coefficient $2^{-k-1}d_1 = p_k$. Multiply it by the Stirling expansion (5.7). A finite Taylor remainder for the rational ratio and the finite-order Stirling remainder give the stated error for each fixed J . Applying (5.5) gives (5.10); the derivation is also checked symbolically in the supplied program. $\square$

Both natural meanings of the OEIS growth-rate comment now follow:

$$\lim_{n \rightarrow \infty} T(n, k)^{1/n} = 4, \quad \lim_{n \rightarrow \infty} \frac{T(n+1, k)}{T(n, k)} = 4 \quad (k \geq 2). \quad (5.11)$$

For reference, the corresponding normalized-probability expansion starts

$$\frac{T(n, k)}{C_n} = p_k \left(1 + \frac{r_1(k)}{n} + \frac{r_2(k)}{n^2} + O_k(n^{-3}) \right), \quad (5.12)$$

where

$$\begin{aligned} r_1(k) &= -\frac{k(k^2 - k - 26)}{8(k+4)}, \\ r_2(k) &= \frac{k(k^4 - 8k^3 - 19k^2 + 158k - 408)}{96(k+4)}. \end{aligned} \quad (5.13)$$

The dependence on k already warns that these fixed-column expansions cannot be used uniformly all the way across the triangle.

6 The discrete limiting law and an exponentially weighted error

Theorem 6.1. *Let K_n be defined under the uniform measure on $\text{Av}_n(123)$. It converges in distribution to K with probabilities p_k in (1.6). Its probability generating function, mean, variance, and tail are*

$$\begin{aligned} P(y) &= \mathbb{E}y^K = \frac{y^2(3-y)}{2(2-y)^3}, & |y| < 2, \\ \mathbb{E}K &= \frac{9}{2}, \quad \text{Var}(K) = \frac{21}{4}, \\ \mathbb{P}(K \geq m) &= 2^{-m-2}(m^2 + 5m + 2), & m \geq 1. \end{aligned} \quad (6.1)$$

For every fixed $1 < r < 2$,

$$\sum_{k \geq 1} r^k |\mathbb{P}(K_n = k) - p_k| = O_r(n^{-1}), \quad (6.2)$$

where $p_1 = 0$ and the finite- n probabilities are zero for $k > n$. Consequently all moments converge, with error $O(n^{-1})$ for each fixed moment.

Proof. Summing the quadratic times geometric expression for p_k gives $P(y)$, with $P(1) = 1$. Derivatives at $y = 1$ give the two moments. Summing the same geometric tail gives the last formula in (6.1).

For the error assertion, set $z = \sqrt{1 - 4x}$ in the exact generating function. It becomes

$$F(x, y) = -\frac{y(z-1)(yz^2 - yz + z + 1)}{(yz - y + 2)(yz^2 - y + 2)}. \quad (6.3)$$

Fix $R < 2$. Uniformly for $|y| \leq R$, neither denominator in (1.4) vanishes for $|x| \leq 1/4$: indeed $|2xy| \leq R/2 < 1$ and $|xyC(x)| \leq R/2 < 1$. By compactness the function admits a common slit, dented neighborhood of its dominant square-root singularity. Its expansion at $z = 0$ has coefficient $-2P(y)$ on z . The next nonanalytic term is of order z^3 ; retaining further terms gives uniform coefficient remainders. Standard square-root coefficient transfer [4, Chapter VI] therefore gives

$$\sup_{|y| \leq R} \left| \frac{Z_n(y)}{C_n} - P(y) \right| = O_R(n^{-1}). \quad (6.4)$$

One may also obtain the transfer constants from $[x^n](1-4x)^\alpha = 4^n \Gamma(n-\alpha)/(\Gamma(-\alpha)\Gamma(n+1))$ for nonintegral α ; the uniform analytic remainder is handled on the same dented contour.

Now choose $r < R < 2$. Cauchy's coefficient bound applied in the variable y to (6.4) gives $|\mathbb{P}(K_n = k) - p_k| \leq M_R n^{-1} R^{-k}$. Sum the convergent geometric majorant $(r/R)^k$ to prove (6.2). Polynomial moment weights are bounded by a constant times r^k , which proves moment convergence and its rate. $\square$

The exponential weight is useful: an unweighted pointwise limit alone would not justify interchanging arbitrary moment sums and limits. Here it is justified with a quantitative bound.

7 The square-root crossover and uniform moving-column regimes

7.1 An exact positive factorization

For $n > k \geq 2$, put $\ell = n - k$ and write

$$\begin{aligned} A_{n,k} &= 2^{k-1} \binom{2\ell}{\ell-1} = 2^{k-1} \ell C_\ell, \\ B_{n,k} &= \binom{2\ell+k-1}{\ell-1}, \quad T(n,k) = A_{n,k} - B_{n,k} = A_{n,k}(1 - R_{n,k}). \end{aligned} \quad (7.1)$$

The ratio has the exact product

$$R_{n,k} = \frac{B_{n,k}}{A_{n,k}} = \prod_{j=1}^{k-1} \frac{1 + j/(2\ell)}{1 + (j+1)/\ell} = \prod_{j=1}^{k-1} \left(1 - \frac{j+2}{2(\ell+j+1)} \right). \quad (7.2)$$

All factors lie strictly between zero and one.

Lemma 7.1. *For every $n > k \geq 2$,*

$$0 < R_{n,k} \leq \exp \left(-\frac{(k-1)(k+4)}{4n} \right). \quad (7.3)$$

If $k/\ell \rightarrow 0$, then

$$\log R_{n,k} = -\frac{(k-1)(k+4)}{4\ell} + O\left(\frac{k^3}{\ell^2}\right). \quad (7.4)$$

The last error is uniform when k/ℓ is sufficiently small.

Proof. Use $\log(1-u) \leq -u$ in (7.2), the bound $\ell + j + 1 \leq n$, and $\sum_{j=1}^{k-1} (j+2) = (k-1)(k+4)/2$. For (7.4), expand $\log(1 + j/(2\ell)) - \log(1 + (j+1)/\ell)$. The first-order terms sum to the displayed expression; the sum of the quadratic remainder bounds is $O(k^3/\ell^2)$. $\square$

Theorem 7.2 (Three uniform regimes). *The following assertions hold.*

(i) *If $2 \leq k = o(\sqrt{n})$, then uniformly in this regime,*

$$\frac{T(n,k)}{C_n} = p_k \left(1 + O\left(\frac{k^2}{n}\right) \right). \quad (7.5)$$

(ii) *If $k/\sqrt{n} \rightarrow s \in (0, \infty)$, then*

$$T(n,k) \sim \frac{4^n 2^{-k}}{2\sqrt{\pi n}} \left(1 - e^{-s^2/4} \right). \quad (7.6)$$

Equivalently,

$$\frac{T(n,k)/C_n}{p_k} \longrightarrow \frac{4(1 - e^{-s^2/4})}{s^2}. \quad (7.7)$$

The equivalents are uniform on compact subintervals of $0 < s < \infty$.

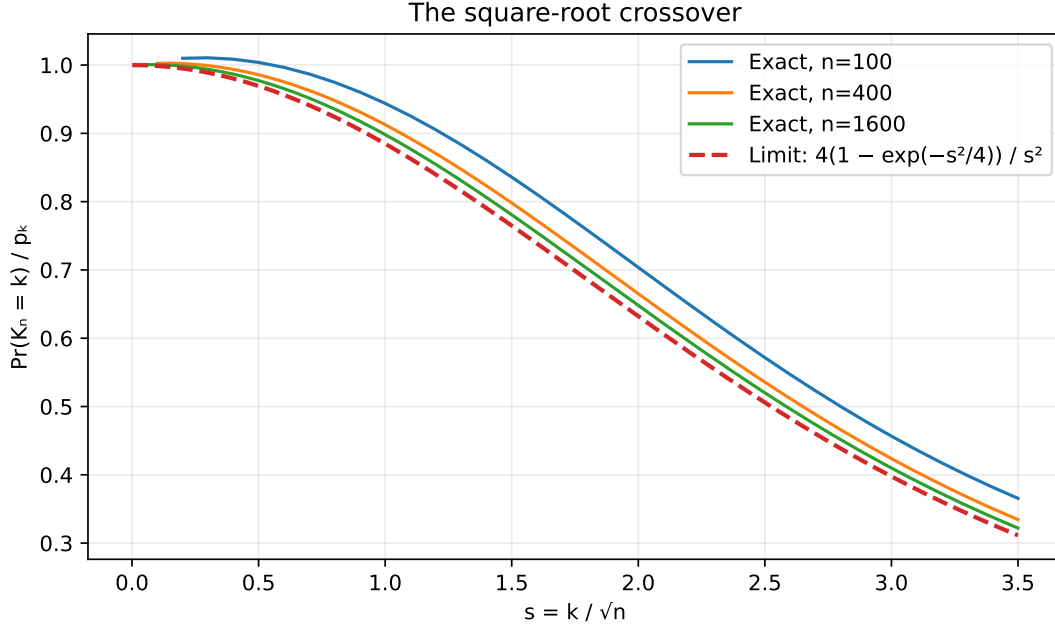


Figure 1: Exact normalized column probabilities approach the crossover function in (7.7). The horizontal coordinate is $k/\sqrt{n}$. These curves show why a fixed- k expansion must not be used uniformly across the whole triangle.

(iii) If $k/\sqrt{n} \rightarrow \infty$ and $\ell = n - k \rightarrow \infty$, then for each fixed J ,

$$T(n, k) = \frac{4^n 2^{-k}}{2\sqrt{\pi}\ell} \left(\sum_{j=0}^J \frac{\gamma_j}{\ell^j} + O_J(\ell^{-J-1}) \right) (1 - R_{n,k}), \quad (7.8)$$

where $\gamma_0 = 1$, $\gamma_1 = -9/8$, $\gamma_2 = 145/128$, and the exponentially small ratio is bounded by (7.3).

Proof. The factor $A_{n,k}$ has the uniform Catalan expansion

$$A_{n,k} = \frac{4^n 2^{-k}}{2\sqrt{\pi}\ell} \left(1 - \frac{9}{8\ell} + O(\ell^{-2}) \right).$$

In the first regime, (7.4) and the expansion of $1 - e^{-u}$ give

$$1 - R_{n,k} = \frac{(k-1)(k+4)}{4\ell} \left(1 + O(k^2/n) \right).$$

Dividing by C_n and using $\ell/n = 1 - k/n$ proves (7.5). In the second regime $\ell/n \rightarrow 1$ and $R_{n,k} \rightarrow e^{-s^2/4}$ uniformly on the stated compact sets. This proves (7.6), and comparison with $p_k \sim k^2/2^{k+3}$ proves (7.7). In the third regime, the bound on $R_{n,k}$ tends to zero while the Catalan expansion at ℓ can be taken to any fixed order. Retaining $1 - R_{n,k}$ exactly yields the stated error without confusing algebraic and exponential orders. $\square$

8 Two exponential sectors and the near-diagonal regime

8.1 Linear rays

Suppose $\beta = (n - k)/n$ remains in a compact subset of $(0, 1)$. Define

$$\begin{aligned} h_0(\beta) &= (1 + \beta) \log 2, \\ h_1(\beta) &= (1 + \beta) \log(1 + \beta) - \beta \log \beta, \\ J(\beta) &= h_0(\beta) - h_1(\beta) > 0. \end{aligned} \quad (8.1)$$

The exact decomposition (7.1) gives two separately expandable exponential sectors. If $C(t) = e^{L_C(t)}$ denotes the formal Catalan correction series, then

$$A_{n,k} \sim \frac{e^{nh_0(\beta)}}{2\sqrt{\pi\beta n}} C\left(\frac{1}{\beta n}\right), \quad (8.2)$$

$$B_{n,k} \sim \sqrt{\frac{\beta}{2\pi(1+\beta)n}} e^{nh_1(\beta)} e^{L_\beta(1/n)}, \quad (8.3)$$

where

$$L_\beta(t) = \sum_{r \geq 1} \frac{B_{2r}\{(1+\beta)^{1-2r} - \beta^{1-2r} - 1\}}{2r(2r-1)} t^{2r-1}. \quad (8.4)$$

These follow directly from Stirling's formula, using $B_{n,k} = \frac{\beta}{1+\beta} \binom{(1+\beta)n}{\beta n}$. At every finite order the relative remainders are uniform on the stated compact sets.

In particular,

$$R_{n,k} \sim \frac{\sqrt{2}\beta}{\sqrt{1+\beta}} e^{-nJ(\beta)}. \quad (8.5)$$

To see $J > 0$, note that $J(1) = 0$ and $J'(\beta) = \log(2\beta/(1+\beta)) < 0$ for $0 < \beta < 1$. Near the endpoint,

$$J(\beta) = \frac{(1-\beta)^2}{4} + O((1-\beta)^3), \quad (8.6)$$

which explains the square-root crossover from the exponential-sector side.

Remark 8.1 (What an exponentially small term does and does not mean). Equations (8.2)–(8.3), together with the exact identity $T = A - B$, provide a two-sector asymptotic description. A finite inverse-power approximation to A has an error much larger than B . Thus it would be incorrect to append the leading B term to an arbitrarily truncated A series and claim the remainder is smaller than B . To resolve the second sector numerically, retain A exactly through its binomial or gamma expression, or control its approximation to exponentially small accuracy. Here both sectors are defined exactly before either is expanded; no resurgence or optimal-truncation theorem is being assumed.

8.2 Fixed distance from the diagonal

For an integer $\ell \geq 1$ fixed and $n > \ell$, the exact formula specializes to

$$T(n, n - \ell) = 2^{n-\ell-1} \left(\binom{2\ell}{\ell-1} - \binom{n+\ell-1}{\ell-1} \right). \quad (8.7)$$

The second term is a polynomial in n of degree $\ell - 1$. This is a literal exponential-minus-polynomial expression, not just an asymptotic equivalent. For example,

$$T(n, n-1) = 2^{n-2} - 1, \quad T(n, n-2) = 2^{n-1} - (n+1).$$

The diagonal itself is 2^{n-1} . The formulas are valid with their stated ranges; coincidences with the first zero column at small n are handled without extra exceptions.

9 Critical weighting: exact normalization and a beta law

9.1 An exact critical partition function

For $y > 0$ define a tilted measure by

$$\mathbb{P}_{n,y}(K_n = k) = \frac{T(n, k)y^k}{Z_n(y)}. \quad (9.1)$$

The measure at $y = 1$ is uniform on $\text{Av}_n(123)$. The value $y = 2$ exactly balances the factor 2^{-k} visible in the asymptotics, so it requires a separate analysis.

Theorem 9.1 (Critical normalization and mean). *For $n \geq 1$,*

$$Z_n(2) = (n+2) \binom{2n}{n} - 4^n, \quad (9.2)$$

$$\mathbb{E}_{n,2} \frac{K_n}{n} = \frac{2}{3} + \frac{4^n}{6Z_n(2)}. \quad (9.3)$$

In particular,

$$Z_n(2) = \frac{4^n \sqrt{n}}{\sqrt{\pi}} \left(1 - \frac{\sqrt{\pi}}{\sqrt{n}} + \frac{15}{8n} - \frac{31}{128n^2} + \frac{21}{1024n^3} + O(n^{-4}) \right), \quad (9.4)$$

$$\mathbb{E}_{n,2}(K_n/n) = \frac{2}{3} + \frac{\sqrt{\pi}}{6\sqrt{n}} + O(n^{-1}). \quad (9.5)$$

The partition function has a full expansion obtained by continuing the central-binomial Stirling series in (9.2).

Proof. In terms of $z = \sqrt{1-4x}$, (6.3) at $y = 2$ reduces to

$$F(x, 2) = -1 + \frac{3}{2z} - \frac{1}{z^2} + \frac{1}{2z^3}. \quad (9.6)$$

Use

$$[x^n]z^{-1} = \binom{2n}{n}, \quad [x^n]z^{-2} = 4^n, \quad [x^n]z^{-3} = (2n+1) \binom{2n}{n}$$

for $n \geq 1$ to obtain (9.2). Differentiating the exact bivariate function gives

$$y \frac{\partial F}{\partial y} \Big|_{y=2} = -\frac{1}{2z} + \frac{1}{2z^2} - \frac{1}{2z^4} + \frac{1}{2z^5}. \quad (9.7)$$

Since $[x^n]z^{-4} = (n+1)4^n$ and $[x^n]z^{-5} = \frac{(2n+1)(2n+3)}{3} \binom{2n}{n}$, coefficient extraction gives

$$\sum_{k=1}^n k 2^k T(n, k) = \frac{2n(n+2)}{3} \binom{2n}{n} - \frac{n}{2} 4^n.$$

Divide by $nZ_n(2)$ and rearrange to prove (9.3). Finally, substitute

$$\binom{2n}{n} = \frac{4^n}{\sqrt{\pi n}} \left(1 - \frac{1}{8n} + \frac{1}{128n^2} + \frac{5}{1024n^3} + O(n^{-4}) \right)$$

in the exact expressions. The term -4^n is responsible for the half-power correction in (9.4). $\square$

9.2 An elementary positive-measure comparison

To study the entire distribution, use the same binomial expressions in (7.1) also for $k = 1 < n$, where $A_{n,1} = B_{n,1}$. Extend the positive first term to the diagonal by $A_{n,n} = 2^{n-1}$, and put $B_{n,n} = 0$. Set

$$b_0 = 1, \quad b_\ell = 4^{-\ell} \binom{2\ell}{\ell-1} \quad (\ell \geq 1). \quad (9.8)$$

Writing $\ell = n - k$ gives

$$2^k A_{n,k} = \frac{4^n}{2} b_\ell, \quad 0 \leq \ell \leq n-1. \quad (9.9)$$

These positive weights are close, after normalization, to the actual critical weights $2^k T(n, k)$.

Lemma 9.2. *For $N \geq 0$,*

$$\sum_{\ell=0}^N b_\ell = -1 + \frac{(2N+1)(N+2)}{N+1} \frac{\binom{2N}{N}}{4^N} = \frac{2\sqrt{N}}{\sqrt{\pi}} - 1 + O((N+1)^{-1/2}), \quad (9.10)$$

where the last estimate is interpreted with a bounded error also at $N = 0$. Moreover,

$$\begin{aligned} \sum_{k=1}^n 2^k A_{n,k} &= (n+1) \binom{2n}{n} - \frac{4^n}{2}, \\ \sum_{k=1}^n 2^k B_{n,k} &= \frac{4^n}{2} - \binom{2n}{n}. \end{aligned} \quad (9.11)$$

Proof. The finite-sum identity in (9.10) is verified at $N = 0$ and then by subtracting its values at N and $N - 1$, using the ratio of successive central binomial coefficients. Stirling's formula gives the estimate. Multiply the identity with $N = n - 1$ by $4^n/2$ and simplify to get the first equation in (9.11). Subtract (9.2) to get the second. $\square$

Theorem 9.3 (Critical beta limit and error). *Under $\mathbb{P}_{n,2}$,*

$$\frac{K_n}{n} \Rightarrow X, \quad X \sim \text{Beta}(1, \tfrac{1}{2}), \quad f_X(t) = \frac{1}{2\sqrt{1-t}} \quad (0 < t < 1). \quad (9.12)$$

Writing $F_X(t) = 1 - \sqrt{1-t}$ on $[0, 1]$, the distribution functions obey

$$\sup_{0 \leq t \leq 1} |\mathbb{P}_{n,2}(K_n/n \leq t) - F_X(t)| = O(n^{-1/2}). \quad (9.13)$$

All moments converge; in particular the limiting mean is $2/3$ and the limiting variance is $4/45$.

Proof. Let $\tilde{\mathbb{P}}_n$ be the measure on $k = 1, \dots, n$ proportional to $2^k A_{n,k}$. If $A = \sum 2^k A_{n,k}$ and $B = \sum 2^k B_{n,k}$, then $Z_n(2) = A - B$. Since all subtracted weights are nonnegative,

$$\sum_k \left| \frac{2^k T(n, k)}{A - B} - \frac{2^k A_{n,k}}{A} \right| \leq \frac{2B}{A} = O(n^{-1/2}),$$

by (9.11). Thus the two measures differ in total variation by $O(n^{-1/2})$.

Under $\tilde{\mathbb{P}}_n$, the deficit $\ell = n - k$ has weights b_ℓ . The exact partial sum (9.10) and its error imply, uniformly for $0 \leq u \leq 1$,

$$\tilde{\mathbb{P}}_n\{(n - K_n)/n \leq u\} = \sqrt{u} + O(n^{-1/2}).$$

Indeed the numerator is a partial sum through an index within one of un , the denominator is the sum through $n - 1$, and the estimates in (9.10) give the claimed uniform error, including both endpoints. Taking complements and adjusting a possible lattice atom (which is also $O(n^{-1/2})$) gives the beta law and (9.13). Since $K_n/n \in [0, 1]$, weak convergence implies convergence of all moments. The beta density directly gives the stated mean and variance. $\square$

Proposition 9.4 (The first bulk correction). *Uniformly for t in any compact subinterval of $(0, 1)$,*

$$\mathbb{P}_{n,2}(K_n/n \leq t) = F_X(t) + \frac{\sqrt{\pi}}{\sqrt{n}} \left(F_X(t) - \frac{1}{2} \right) + O(n^{-1}). \quad (9.14)$$

This expansion is not asserted uniformly through the endpoints.

Proof. Put $m = \lfloor nt \rfloor$. The positive cumulative weight is $\frac{4^n}{2} \sum_{\ell=n-m}^{n-1} b_\ell$. The constant -1 cancels in the difference of the two partial sums in (9.10). Since both endpoints are proportional to n ,

$$\sum_{k \leq m} 2^k A_{n,k} = \frac{4^n \sqrt{n}}{\sqrt{\pi}} F_X(t) + O(4^n n^{-1/2}).$$

For $k > m$, the ratio bound (7.3) makes the total B weight exponentially small compared with $4^n \sqrt{n}$. Hence

$$\sum_{k \leq m} 2^k B_{n,k} = \frac{4^n}{2} - \binom{2n}{n} + O(4^n e^{-cn})$$

for some $c > 0$ depending on the compact interval (a polynomial factor can be absorbed by reducing c). The numerator is therefore $\frac{4^n \sqrt{n}}{\sqrt{\pi}} F_X(t) - 4^n/2 + O(4^n n^{-1/2})$. Divide by (9.4) to obtain the result. The floor in m changes the leading cumulative expression by only $O(n^{-1})$ after normalization on the stated compact sets. $\square$

10 The full tilted phase diagram and its critical window

10.1 Below the transition

Theorem 10.1 (Subcritical law). *For every fixed $0 < y < 2$,*

$$Z_n(y) = P(y) C_n (1 + O_y(n^{-1})), \quad \mathbb{P}_{n,y}(K_n = k) \longrightarrow \frac{p_k y^k}{P(y)}. \quad (10.1)$$

The limiting law has an exponential moment, and $K_n/n \rightarrow 0$ in probability. The partition function has an all-orders inverse-power expansion at each fixed subcritical y .

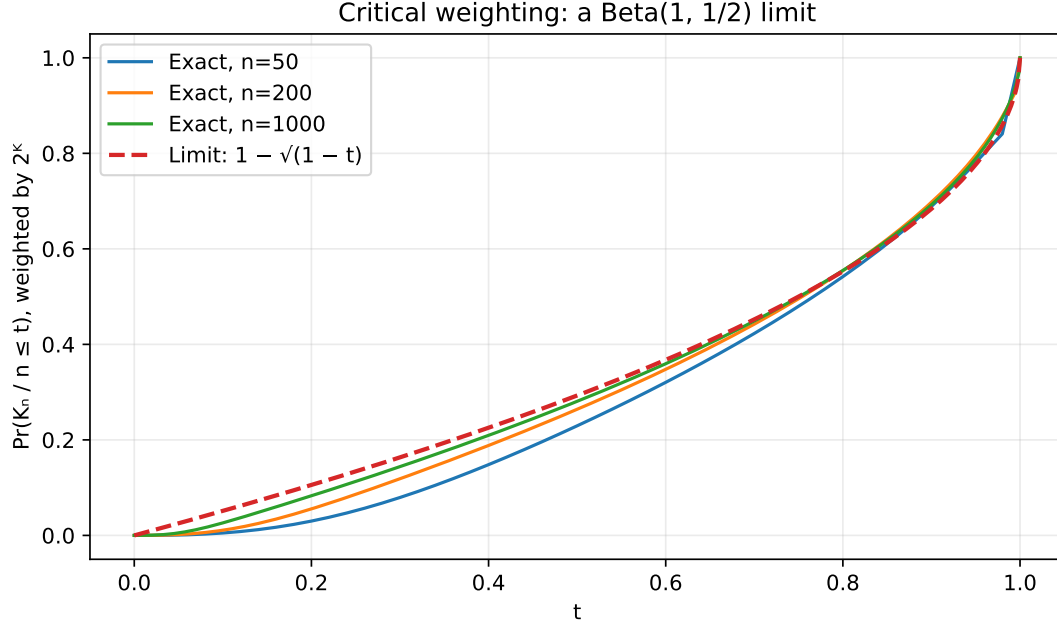


Figure 2: At the critical weight 2^{K_n} , the cumulative distribution of K_n/n approaches $1 - \sqrt{1-t}$. Convergence is only of order $n^{-1/2}$; the exact mean in (9.3) quantifies one part of this finite-size effect.

Proof. The uniform generating-function estimate (6.4) gives the normalization. Combining it with the fixed-column probabilities gives the pointwise limit, whose sum is one by definition of P . Choose a circle $|u| = R < 2$ with $R > y$. The same coefficient estimates give exponentially weighted convergence after multiplying by y^k , so the family is tight and has uniformly bounded moments. Expanding (6.3) to further odd powers of z gives every fixed inverse-power order of the partition function by the same transfer argument. $\square$

[write] Added 5 October 2026, batch 98. Part II's Theorem 17.2 is this theorem for every pattern $1(r+1)r \cdots 2$ (with 2 replaced by $y_c = \sec^2(\pi/(r+2))$); at $r = 2$ its function L_2 is P . Part II also writes the limit law as an independent sum of geometric variables (Theorem 17.3 and the note after it). At $y = y_c$ and $r \geq 3$ the critical behaviour differs from Theorem 9.3 (Theorem 20.1).

10.2 Above the transition

For $y > 2$ put

$$q = \sqrt{1 - 2/y}, \quad \mathcal{A}(y) = \frac{1}{2} \left(1 + \frac{1}{q} - C\left(\frac{1}{2y}\right) \right) = \frac{q^2 + 1}{2q(q+1)}. \quad (10.2)$$

Theorem 10.2 (Supercritical law). *For every fixed $y > 2$, there is $0 < \eta_y < 1$ such that*

$$Z_n(y) = \mathcal{A}(y)(2y)^n (1 + O_y(\eta_y^n)). \quad (10.3)$$

The deficit $D_n = n - K_n$ converges in distribution to a nonnegative integer-valued random variable D_y with

$$\begin{aligned} \mathbb{P}(D_y = 0) &= \frac{1}{2\mathcal{A}(y)}, \\ \mathbb{P}(D_y = \ell) &= \frac{\binom{2\ell}{\ell-1}}{2\mathcal{A}(y)(2y)^\ell}, \quad \ell \geq 1. \end{aligned} \quad (10.4)$$

Its generating function is

$$\mathbb{E}u^{D_y} = \frac{H(u/(2y))}{H(1/(2y))}, \quad H(t) = 1 + (1 - 4t)^{-1/2} - C(t). \quad (10.5)$$

In particular, $K_n/n \rightarrow 1$ in probability.

Proof. The pole $\rho = 1/(2y)$ of $1/(1 - 2xy)$ lies strictly inside the Catalan radius $1/4$. The other denominator in (1.4) is nonzero at ρ , since $\rho y C(\rho) = C(\rho)/2 < 1$. In fact it has no zero in a slightly larger disk: the bound $|xyC(x)| \leq |x|yC(|x|)$ and continuity suffice. The numerator of the simple pole is

$$\lim_{x \rightarrow \rho} (1 - 2yx)F(x, y) = \frac{1}{2} + \frac{C(\rho)^3}{8y\{1 - C(\rho)/2\}} = \mathcal{A}(y).$$

Subtract $\mathcal{A}(y)/(1 - 2yx)$ from F . The remainder is analytic in a disk of radius strictly larger than ρ , so Cauchy's coefficient bound proves (10.3).

For fixed $\ell \geq 1$, multiply (8.7) by $y^{n-\ell}$ and divide by (10.3). The polynomial term tends to zero relative to the exponential term, and the limit is (10.4). The diagonal gives the mass at zero. Finally,

$$\sum_{\ell \geq 1} \binom{2\ell}{\ell-1} t^\ell = (1 - 4t)^{-1/2} - C(t),$$

so the limiting masses sum to $H(1/(2y))/(2\mathcal{A}(y)) = 1$. This proves tightness, the asserted law, and its generating function. $\square$

The limiting deficit has an exponential tail since $1/(2y) < 1/4$. At fixed $y > 2$, moments converge as well: for $u > 1$ sufficiently close to one, $y/u > 2$ and

$$\mathbb{E}_{n,y} u^{D_n} = u^n \frac{Z_n(y/u)}{Z_n(y)} \rightarrow \frac{\mathcal{A}(y/u)}{\mathcal{A}(y)} = \frac{H(u/(2y))}{H(1/(2y))}$$

uniformly in a neighborhood of $u = 1$.

10.3 Free energy and a finite-size window

Corollary 10.3. *The limiting exponential rate is*

$$f(y) := \lim_{n \rightarrow \infty} \frac{1}{n} \log Z_n(y) = \begin{cases} \log 4, & 0 < y \leq 2, \\ \log(2y), & y \geq 2. \end{cases} \quad (10.6)$$

As a function of $\log y$, its one-sided derivatives at $y = 2$ are 0 and 1. At exactly $y = 2$, the finite-size mean fraction instead converges to $2/3$, in accordance with the nondegenerate beta law.

This derivative jump is the precise sense in which the model has a first-order transition. It is a description of the proved weighted combinatorial model, not a claim about a physical system.

Theorem 10.4 (Critical window). *Let $\lambda \in \mathbb{R}$ be fixed and $y_n = 2e^{\lambda/n}$. Then*

$$\begin{aligned} \frac{Z_n(y_n)}{Z_n(2)} &= M(\lambda) + O_\lambda(n^{-1/2}), \\ M(\lambda) &= \int_0^1 \frac{e^{\lambda t}}{2\sqrt{1-t}} dt = e^\lambda \int_0^1 e^{-\lambda u^2} du. \end{aligned} \quad (10.7)$$

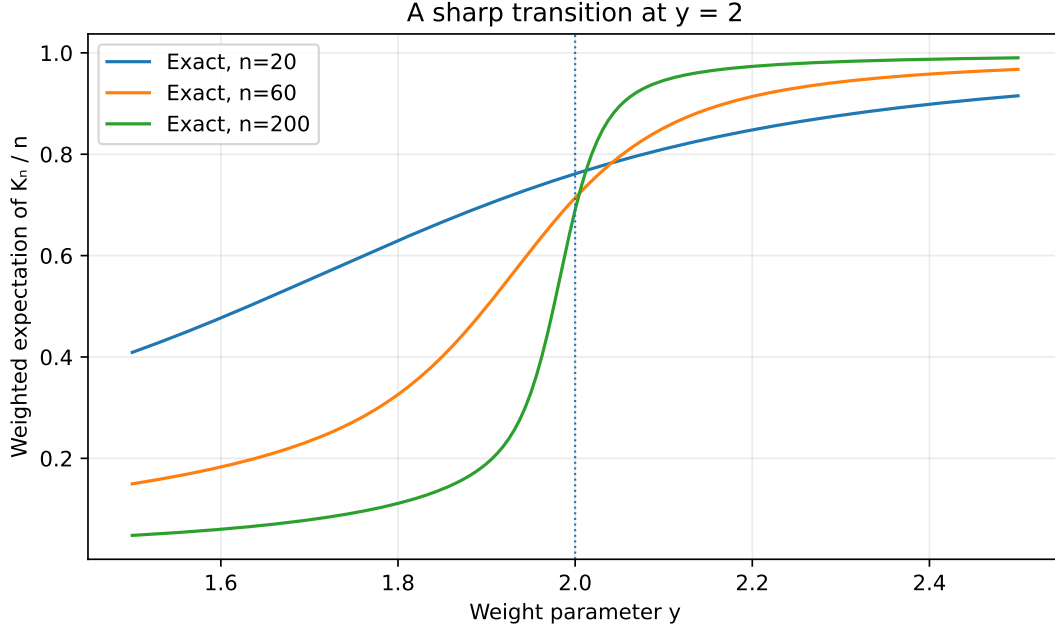


Figure 3: Exact tilted means for increasing n . The limiting mean fraction is 0 for fixed $y < 2$ and 1 for fixed $y > 2$, with the distinct critical value $2/3$. The window $y = 2e^{\lambda/n}$ resolves the transition rather than replacing it by a discontinuous finite- n approximation.

Under $\mathbb{P}_{n,y_n}$, K_n/n converges to the density

$$f_\lambda(t) = \frac{e^{\lambda t}}{2M(\lambda)\sqrt{1-t}}, \quad 0 < t < 1. \quad (10.8)$$

Both the normalization error and the distribution-function error are $O(n^{-1/2})$ uniformly for λ in a compact real interval.

Proof. The normalization ratio is exactly $\mathbb{E}_{n,2} \exp(\lambda K_n/n)$. Apply Theorem 9.3; integration by parts bounds the difference of expectations by the distribution-function error times the total variation of $e^{\lambda t}$ on $[0, 1]$. This variation is uniformly bounded for λ in a compact interval. The substitution $t = 1 - u^2$ gives the second integral. For the tilted distribution functions, apply the same argument to $e^{\lambda t} \mathbf{1}_{t \leq a}$ and divide by the normalization, which remains uniformly bounded away from zero. This proves the stated limit and rate. $\square$

When $\lambda > 0$ one may write $M(\lambda) = e^\lambda \sqrt{\pi} \operatorname{erf}(\sqrt{\lambda})/(2\sqrt{\lambda})$; the integral form is real and nonsingular for every real λ , with $M(0) = 1$.

11 Asymptotic inversion and integer thresholds

11.1 A natural smooth continuation

A discrete sequence does not have a unique real inverse without choosing an interpolation. For fixed $k \geq 2$ we use the gamma continuation of the exact binomial formula:

$$T_k(x) = \frac{2^{k-1}\Gamma(2x-2k+1)}{\Gamma(x-k)\Gamma(x-k+2)} - \frac{\Gamma(2x-k)}{\Gamma(x-k)\Gamma(x+1)}, \quad x > k. \quad (11.1)$$

It equals $T(n, k)$ at all integers $n > k$. The product formula (7.2), now with $\ell = x - k > 0$, proves positivity because each factor is less than one. Differentiated Stirling expansions show that

$$\log T_k(x) \sim ax - b \log x + \log A_k^* + \sum_{j \geq 1} \ell_j(k) x^{-j}, \quad (11.2)$$

as an all-orders asymptotic expansion, where

$$a = \log 4, \quad b = \frac{3}{2}, \quad A_k^* = \frac{p_k}{\sqrt{\pi}}, \quad \ell_1 = c_1, \quad \ell_2 = c_2 - \frac{1}{2}c_1^2.$$

Its derivative is $a - b/x + O(x^{-2}) > 0$ for all sufficiently large x . Thus sufficiently large targets have a unique inverse on the eventual increasing branch.

11.2 Lambert- W reduction and corrections

For a target $v \rightarrow \infty$, put

$$Y = \log(v/A_k^*), \quad N = -\frac{b}{a} W_{-1} \left(-\frac{a}{b} e^{-Y/b} \right). \quad (11.3)$$

The real branch W_{-1} , not the principal branch, is required to make $N \rightarrow \infty$. The defining identity for W implies $aN - b \log N = Y$; standard branch conventions are recorded in DLMF §4.13 [6].

Theorem 11.1 (Inverse expansion). *Let $x(v)$ be the eventual increasing inverse of T_k . Then*

$$x(v) = N - \frac{\ell_1}{aN} - \frac{\ell_2/a + b\ell_1/a^2}{N^2} + O_k(N^{-3}). \quad (11.4)$$

An expansion to every order is determined recursively by substituting $\delta = \sum_{j \geq 1} d_j N^{-j}$ in

$$a\delta - b \log(1 + \delta/N) + \sum_{j \geq 1} \ell_j(N + \delta)^{-j} = 0 \quad (11.5)$$

and setting each coefficient equal to zero. At order N^{-j} , the new unknown d_j has coefficient $a \neq 0$.

Proof. Write $x = N + \delta$ in (11.2) and use the exact equation for N . The remaining equation is (11.5). At its first two orders,

$$ad_1 + \ell_1 = 0, \quad ad_2 - bd_1 + \ell_2 = 0,$$

which gives (11.4). Inductively, after the first J coefficients are chosen, the residual in the logarithmic equation is $O(N^{-J-1})$. Its derivative in δ is $a + O(N^{-1})$, bounded away from zero. The mean value theorem transfers the residual to an inverse error of the same order. Differentiated Stirling remainder bounds justify the derivatives used here [5]. $\square$

[write] **Added 1 October 2026, batch 74: an instance of repository results.** Theorem 11.1 is the balanced inverse theorem `t2:thm:balanced-inverse` of the combinatorial-inverses volume [7], applied to the gamma continuation (11.1) (a gamma quotient of the kind covered by that volume's engine `t2:thm:gamma`) with $a = \log 4$, $\beta = -b = -\frac{3}{2}$, $C = A_k^*$ and $q_j = \ell_j(k)$. That theorem's $d_1 = -q_1/a$ and $d_2 = \beta q_1/a^2 - q_2/a$ are the two corrections in (11.4); it also gives a nonrecursive formula for every coefficient, and its branch rule (W_{-1} for $\beta < 0$) is the one used here. The integer-threshold identity $Q_k(v) = \lceil x(v) \rceil$ and the bracket (11.7) below are parts (1) and (2) (exact rounding and the separation condition) of the staircase theorem `p0:thm:staircase` of the transseries-and-inversion volume [8]. The tail inversion (11.9)–(11.10) is a Lambert core of the same kind. No novelty is claimed for this section.

The Lambert quantity itself begins

$$N = \frac{Y + b \log Y - b \log a}{a} + O\left(\frac{\log Y}{Y}\right). \quad (11.6)$$

Repeated substitution, or the standard expansion of W_{-1} , expands it in powers of Y^{-1} with polynomial coefficients in $\log Y$. Keeping N in Lambert form often gives the more accurate practical approximation at a fixed number of displayed terms.

11.3 The rounding issue

Define the eventual integer threshold

$$Q_k(v) = \min\{n \geq k + 1 : T(n, k) \geq v\}.$$

For sufficiently large v , all earlier finitely many values are below the target and the continuation is increasing, so $Q_k(v) = \lceil x(v) \rceil$. However, taking the ceiling of a *truncated* inverse expansion is not uniformly safe: the exact inverse may lie arbitrarily close to an integer. If a rigorous remainder bound is $|x(v) - x_J(v)| \leq \varepsilon_J$, the valid conclusion is the bracket

$$\lceil x_J(v) - \varepsilon_J \rceil \leq Q_k(v) \leq \lceil x_J(v) + \varepsilon_J \rceil. \quad (11.7)$$

An exact binomial evaluation resolves any remaining one- or two-integer ambiguity. This report proves asymptotic remainder orders; it does not claim that the floating-point demonstrations alone provide explicit certified remainder constants or effective monotonicity cutoffs.

11.4 Inverting the limiting tail

The limiting discrete law also has a useful inverse asymptotic. Extend its exact tail formula to large real m and solve

$$\varepsilon = 2^{-m-2}(m^2 + 5m + 2), \quad \varepsilon \downarrow 0.$$

Put $u = m + 5/2$ and $a_2 = \log 2$. Then

$$\varepsilon = \sqrt{2} \left(u^2 - \frac{17}{4} \right) e^{-a_2 u}. \quad (11.8)$$

The leading solution is

$$U = -\frac{2}{a_2} W_{-1} \left(-\frac{a_2}{2} \sqrt{\frac{\varepsilon}{\sqrt{2}}} \right). \quad (11.9)$$

Expanding the logarithm of (11.8) about $u = U$ gives

$$m = U - \frac{5}{2} - \frac{17}{4a_2 U^2} + O(U^{-3}). \quad (11.10)$$

Indeed, the neglected factor contributes $\log(1 - 17/(4U^2)) = -17/(4U^2) + O(U^{-4})$ and the derivative of $2 \log u - a_2 u$ is $-a_2 + 2/u$. This determines both the sign and the coefficient of the correction. Further terms follow recursively. Integer quantiles require the same rounding care as in (11.7). This tail inversion concerns the limiting law; for finite n , the exact cumulative formula (4.3) is available and should be preferred when feasible.

12 Independent checks, numerical behavior, and reproducibility

12.1 Three logically separate checks

The archive contains `code/verify.py`, using integer arithmetic for enumeration and `sympy/mpmath` for symbolic and numerical checks. The executed default run performed the following tests.

1. **Direct permutation enumeration.** Every permutation of sizes 1 through 9 was examined: 409,113 permutations in total. A three-term increasing-subsequence test decides 123 avoidance. A separate triple scan finds the largest minimum value of a 132 occurrence, hence $K = n - \max\{\text{minimum of a 132 occurrence}\}$, with an empty maximum of zero. The resulting rows agree with the binomial formula.
2. **Insertion-state dynamic programming.** An implementation of the suffix production rule, carrying a flag for whether the first failure has occurred, checks every row through $n = 120$: 7,260 triangle cells. It verifies Catalan row sums, all cumulative counts, the exact critical partition function, its first two raw moments, the polynomial factorization, and the first-order column recurrence. This does not enumerate from the closed formula being checked.
3. **Symbolic and high-precision checks.** Symbolic reduction verifies equality with the posted generating function, the critical specialization, the limit generating function, and the first three fixed-column corrections. Numerical tables use 100 decimal digits and stable logarithmic evaluation where cancellation is relevant.

All checks passed in the supplied run. These computations are independent support for the proofs, not a substitute for them and not a proof-assistant kernel certificate.

12.2 Illustrative numerical results

Table 1 reports relative errors $\text{approximation}/T(n, k) - 1$. The three approximations retain zero, one, or two inverse-power corrections in (5.8).

Table 2 illustrates the slower critical convergence. The finite mean is always larger than $2/3$ by (9.3); the limiting cumulative probability at $1/2$ is $1 - 1/\sqrt{2} \approx 0.2928932188$.

12.3 Stable evaluation and complexity

The two-binomial expression gives a direct exact evaluation with two binomial coefficients and a power of two. With big integers it is safe from cancellation, although bit complexity grows with the output length. For floating-point evaluation when k is small relative to $\sqrt{n}$, the two leading binomials nearly cancel. The stable form is

$$\log T = \log A + \log(-\text{expm1}(\log R)), \quad (12.1)$$

using `log1p` for the factors in (7.2). The verifier uses this form for the gamma continuation and inverse tests.

k	n	Leading term	Through n^{-1}	Through n^{-2}
2	100	$1.3274 \cdot 10^{-3}$	$7.5743 \cdot 10^{-5}$	$1.4262 \cdot 10^{-6}$
3	100	$6.5417 \cdot 10^{-4}$	$1.1810 \cdot 10^{-4}$	$3.6291 \cdot 10^{-6}$
5	100	$7.3479 \cdot 10^{-3}$	$2.1253 \cdot 10^{-4}$	$6.6022 \cdot 10^{-6}$
8	100	$3.7432 \cdot 10^{-2}$	$-1.7498 \cdot 10^{-4}$	$-2.5042 \cdot 10^{-5}$
2	500	$2.5304 \cdot 10^{-4}$	$2.9808 \cdot 10^{-6}$	$1.1250 \cdot 10^{-8}$
3	500	$1.1176 \cdot 10^{-4}$	$4.6048 \cdot 10^{-6}$	$2.8396 \cdot 10^{-8}$
5	500	$1.4269 \cdot 10^{-3}$	$8.2393 \cdot 10^{-6}$	$5.0570 \cdot 10^{-8}$
8	500	$7.2969 \cdot 10^{-3}$	$-6.0141 \cdot 10^{-6}$	$-1.9069 \cdot 10^{-7}$

Table 1: Fixed-column asymptotic errors from the independently evaluated exact formula. All digits shown are rounded from the included CSV file.

n	$\mathbb{E}_{n,2}(K_n/n)$	$\mathbb{P}_{n,2}(K_n/n \leq 1/2)$
20	0.7614630041	0.1888612232
50	0.7197682165	0.2294764254
100	0.7017725312	0.2503403803
200	0.6902952663	0.2640357949
500	0.6809569222	0.2753274754
1000	0.6765433918	0.2807136187
Limit	$2/3$	$1 - 1/\sqrt{2}$

Table 2: Exact critical probabilities, evaluated with arbitrary-precision arithmetic. The substantial finite-size discrepancy is predicted by the $n^{-1/2}$ correction, rather than evidence against the beta limit.

For a whole table, the suffix-state dynamic program has a straightforward $O(N^3)$ arithmetic-operation implementation through size N , using suffix sums to aggregate state transitions. This is an operation count, not a bit-complexity bound; the integer sizes increase with N . For an individual fixed column, the first-order recurrence is usually more convenient after an initial value is known.

12.4 Rebuilding the archive

The source is `article.tex`; the PDF is compiled with standard LaTeX packages and references embedded in the source. The figure sources and CSV data are included. From the archive directory:

```
python -m pip install -r requirements.txt
python code/verify.py
python code/make_figures.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The verifier writes `data/verification_summary.json` and numerical CSV files. The supplied `verification_log.txt` records the executed run. No network access is needed after dependencies have been installed. `OEIS_update.txt` contains suggested entry amendments; they have not been submitted.

[write] Added 1 October 2026, batch 74: the shipped layout. The archive's directory layout is kept, with two moves: its `requirements.txt` is shipped as `data/requirements.txt` and its Makefile as `code/Makefile`; the delivered PDF is not shipped, and `article.pdf` is a build of this text. The commands above therefore still work from the report directory, with `-r data/requirements.txt`; but `verify.py` rewrites `data/verification_summary.json` and the five CSV files, and `make_figures.py` rewrites `figures/`, so run them on a copy of the directory (the README gives the commands). The verifier needs only `mpmath` and `sympy`; `numpy` and `matplotlib` are for the figures.

13 Formalization boundary and further research

13.1 A concrete path into Provelt

The project can be formalized in layers, with a small exact core rather than an immediate attempt to formalize the entire analytic theory. First define the restriction to the largest k entries and prove its monotonicity with respect to pattern containment. Next formalize deletion and insertion of the minimum as inverse operations, and prove the active suffix lemma. Finite counting then gives the safe-prefix distribution and the first-failure convolution. These are the genuinely combinatorial obligations; a formal manipulation of the conjectured generating function alone would not establish that it counts the specified permutations.

The subsequent exact algebra is well delimited: Catalan powers, the binomial telescoper, the entry and tail formulas, and the polynomial-factor recurrence. Each can be checked without trusting floating-point arithmetic. The critical partition function and its moment identities can then be proved by formal-power-series coefficient extraction.

Finally, analytic modules can establish the Stirling expansion, the uniform coefficient-transfer bound, and the probability limits. The repository's generic asymptotic scales and differential-block framework provide a natural home for the formal expansions and inverse recursion [9]. Their existence does not discharge the analytic remainder obligations. The exact binomial split is especially valuable because it lets a formal statement distinguish the two exponential sectors without assuming an unproved summability or resurgent interpretation.

13.2 Proposed research questions

The following questions go beyond results established in this report. They are proposed research directions, not assertions that every one is a previously published open problem.

1. A family of first-failure statistics. Replace 132 by $1(r+1)r \cdots 2$ while retaining global 123 avoidance. At a 123-active site the right-hand suffix decreases, so a new occurrence is created precisely when at least r entries remain to the right. Thus safe sites become $j = 0, \dots, r-1$, subject to $j \leq d$. Determine the corresponding first-failure generating functions, explicit entry formulas where possible, and the critical exponents and limiting laws. The safe-state process is no longer the same binary reset process; its analysis is the substantive new step.

[write] **Added 5 October 2026, batch 98: answered in Part II.** Part II answers this question for every $r \geq 2$. The safe-site rule stated here is (16.6); the safe-state process is solved by the Chebyshev polynomials Δ_r (Lemmas 16.3 and 17.1; the safe counts are classical, by Chow and West). The first-failure generating function is Theorem 16.1, and the entry formula is the finite positive sum (16.4); a two-binomial closed form like (1.3) is not given for $r \geq 3$. The critical weight is $y_c = \sec^2(\pi/(r+2))$, and for every fixed $r \geq 3$ Part II proves the subcritical law (Theorems 17.2 and 17.3), a tight critical deficit with masses $\asymp m^{-3/2}$, total-variation rate $n^{-1/2}$ and moment threshold $\frac{1}{2}$ (Theorems 20.1 and 20.2), the supercritical law (Theorem 21.1) and a displaced window (Theorem 22.2). Uniformity in r and joint limits $r = r(n)$ remain open (Part II's questions 1 and 2).

2. Joint first-failure and suffix statistics. Retain the active suffix length at the first failing insertion and at the final permutation. Derive a multivariate generating function and determine whether the critical beta law admits a useful joint limit with either suffix statistic. The convolution (3.1) retains the failing state j and is a starting point, but the joint limit is not proved here.

[write] **Added 5 October 2026, batch 98.** Still open. Part II's question 7 generalizes it to every r and adds the early-versus-late phase indicator.

3. A fully uniform higher-order theory. The leading square-root crossover and the bulk critical correction are proved above. Develop higher-order expansions that remain uniform as $k/\sqrt{n}$ tends to zero or infinity, and match them to the fixed-deficit region. Similarly resolve the endpoint layers of the critical cumulative law, where (9.14) is not uniform. Explicit remainder constants would make these expansions directly useful for certified computation.

4. Complex zeros and finite-size scaling. Study the zeros of the polynomial $Z_n(y)$ near $y = 2$ and globally. The critical window suggests that the entire function $M(\lambda) = e^\lambda \int_0^1 e^{-\lambda u^2} du$ should govern local complex scaling. Establish the requisite complex-uniform convergence, locate its zeros, and connect them to the accumulation of zeros of Z_n . Real-window convergence alone does not supply this theorem.

[write] **Added 5 October 2026, batch 98.** Still open at $r = 2$ (Part II's question 5 restates it). For the patterns $1(r+1)r \cdots 2$ with $r \geq 3$, Part II locates local sequences of zeros through the complex-uniform window limit $2a_r + P_r e^{-s}$ (Theorem 23.1); that function and its window are different from $M(\lambda)$ here.

5. The polynomial factors H_k . Determine the geometry and asymptotic distribution of the zeros of H_k as $k \rightarrow \infty$, and seek structural coefficient bases or recurrences that explain them. Positivity on $n > k$ and the leading coefficient are proved here; real-rootedness or any broader zero-location claim is not assumed. A useful outcome could be sharper rational bounds on column ratios.

6. Effective inverse certificates. Make the inverse theorems effective by producing explicit monotonicity cutoffs and remainder constants, then combine interval evaluation of W_{-1} and gamma functions with exact binomial checks. The goal is a certified threshold algorithm, with a proof that a short interval contains the answer, rather than an unqualified rounding of an asymptotic formula.

7. Direct bijections to familiar Catalan objects. Translate the first-failure parameter into a natural Dyck-path or ordered-tree statistic, preferably preserving the two low-column interpretations. The insertion tree already supplies an encoding, but a transparent object-level bijection may explain the two-binomial cancellation and the critical beta law without generating functions.

[write] Added 5 October 2026, batch 98. Part II's question 8 asks the same for every r .

8. Kernel-checked enumeration and reusable analytic interfaces. Implement the exact layer in Lean and use it as a test case for a reusable interface between finite combinatorial enumeration, rational generating functions over the Catalan extension, and asymptotic statements with explicit remainders. The distinction between exact algebraic identities, formal transseries, and analytic estimates should remain visible in the theorem signatures and audit output.

[write] Added 5 October 2026, batch 98. Part II's question 10 extends this to the positive formula for every r and to certified sampling; nothing in either Part is formalized.

13.3 Conclusion

The posted conjecture for A273821 is accessible through the first insertion that creates a forbidden pattern. That combinatorial event leads to a positive Catalan convolution, a telescoping binomial formula, and a complete fixed-column asymptotic expansion. It also reveals behavior invisible in the original growth-rate comment: a square-root crossover, a precise separation of exponential sectors, and a weighted transition from bounded first-failure time to bounded final deficit, with a beta law exactly at criticality. The exact formulas, rather than numerical extrapolation, support every asserted asymptotic regime.

Part II

Longer patterns: exact enumeration, critical condensation and a displaced window

14 Provenance, scope and notation of Part II

[Added 5 October 2026, batch 98.] Part II was added in the write of batch 98. It answers the first proposed research question of Part I (Section 13, question 1): it replaces 132 by $\tau_r = 1(r+1)r \cdots 2$ for every $r \geq 2$, keeping global 123 avoidance, and determines the first-failure generating function, a positive coefficient formula, and, for every fixed $r \geq 3$, the subcritical, critical and supercritical laws, a displaced coexistence window and local complex zeros. It is built from one manuscript, printed in full in Sections 15–28. Where it proves a statement of Part I again (the case $r = 2$), its proof is kept, the Part I statement is named, and the overlap is recorded in Table 4. Additions of this write are set as [write] notes; none of them changes a statement of the manuscript.

14.1 Provenance

Part II is manuscript 07 of batch 98 of ProveIt’s incoming-reports intake, the archive `First_Pattern_Failure_Beyond_132.zip` (24 files, 1,044,533 bytes), which arrived in commit 2172df76a and was placed in this report by commit 0fba5167f. Its article is titled *First Pattern Failure Beyond 132: Exact enumeration, critical condensation, and a logarithmic transition window. An extension of OEIS A273821* (23 pages, dated 5 October 2026); its title page reads “An AI-assisted research manuscript”, and its PDF metadata names “ChatGPT” as author. Its pinned repository commit is 0f93381e8932d3aee3d5c2abfb5c98d128d49b9c (4 October 2026, 19:54 PDT), recorded in its article, its delivery README and its `provenance.json`. The directory of this report is unchanged between that commit and the placement commit, so the manuscript read Part I exactly as it stands. Its delivery README describes the text as having “two independent AI proof audits”; no audit file was delivered.

The writing step changed the manuscript only as follows. Every label carries the prefix `fpf:lp:` (the prefix `fpf:` alone would have made eleven labels collide with Part I’s). The manuscript’s `cleverref` references are written with `\ref`, which this report uses; its citations of the host report are references to Part I; and the symbols of Table 5 are renamed. Part I’s macros print two symbols differently: a falling factorial is $(M)_h$ (the manuscript prints $(M)^h$), and a differential is d (the manuscript’s macro adds a thin space). The manuscript’s sections 1–11 are Sections 15–25 (manuscript section n is Section $n + 14$, and its item and equation $n.m$ is $(n + 14).m$); its section 12 is Section 26, and its Appendices A and B are Sections 27 and 28. Part I’s Appendices A and B follow Part II and belong to Part I.

Scope and non-claims of manuscript 07, kept. The unrefined enumeration of permutations avoiding both patterns is classical: Chow and West [10], Theorem 3.1, give the Chebyshev quotient and growth constant; Krattenthaler [11], Theorem 9, gives a Dyck-path formulation, and his Theorem 8 marks occurrences rather than first failure; first-ascent stopping statistics in Catalan permutations predate the manuscript [12]. The $r = 2$ first-failure formula, the beta law and the identity $Z_n(2) = (n + 2)\binom{2n}{n} - 4^n$ are Part I’s and are credited. Its scope ledger (Table 6) “does not assert that no equivalent refinement exists elsewhere”; no worldwide priority is claimed. The theorems of Sections 19–23 fix $r \geq 3$ and their constants carry no claimed uniformity as $r \rightarrow \infty$; Theorem 18.1 is an iterated limit and does not commute the limits; no joint $r = r(n)$ theorem is asserted. The zero theorem is local, for each fixed zero index: it does not order the zeros globally, describe all of them, or settle Part I’s $r = 2$ zero question. Numerical values and root residuals are not interval certificates, the coexistence curves converge slowly, and the proofs do not rely on the computations. The manuscript is not peer reviewed and is not formalized in Lean or Rocq; its research questions are proposals, “not represented as established open problems”.

14.2 What Part II adds, and its second routes

What is new to the repository: everything for $r \geq 3$; the geometric decomposition and the long-pattern limit for every r , including $r = 2$; the uniform two-index estimate; the critical, supercritical, window and zero theorems for $r \geq 3$; the worked example $r = 3$; and the exact-moment calculation. The safe counts and the Chebyshev spectrum are classical (Chow–West), and the diagonal of Theorem 16.1 is the Chow–West safe count.

14.3 Notation

Part I’s $T, Z_n, F, K_n, C, C_n, a_{k,d}$ and D_d are the cases $r = 2$ of Part II’s $T_r, Z_{n,r}, F_r, K_r$, and the same $C, C_n, a_{k,d}$ and D_d . Table 5 lists the symbols of manuscript 07 renamed in Part II. No normalization is changed.

Topic	Part I ($\tau_2 = 132$)	Part II ($\tau_r, r \geq 2$)
Exact counts	$T(n, k)$ as a difference of two binomials (1.3), via the convolution (3.1)	the generating function F_r and the finite positive sum (16.4) for every $r \geq 2$; no two-term closed form for $r \geq 3$
Fixed columns	all-orders expansion (Theorem 5.2)	leading term with growth constant 4 for every r (16.11)
Subcritical weights	$0 < y < 2$ (Theorem 10.1)	$0 < y < y_c$, $y_c = \sec^2(\pi/(r+2))$; an independent geometric sum (Theorem 17.3)
Critical weight	$K_n/n \Rightarrow \text{Beta}(1, \frac{1}{2})$ (Theorem 9.3)	for $r \geq 3$ a tight deficit with mass $\asymp m^{-3/2}$, sharp total-variation rate and the full moment threshold (Theorems 20.1, 20.2)
Supercritical weights	$y > 2$ (Theorem 10.2)	$y > y_c$, $r \geq 3$, as an exponential tilt of the critical law (Theorem 21.1)
Window	$y = 2e^{\lambda/n}$ (Theorem 10.4)	displaced by $(\frac{1}{2} \log n + \log \log n)/n$, two-point mixture (Theorem 22.2)
Complex zeros	open (question 4)	local zeros for $r \geq 3$ (Theorem 23.1); $r = 2$ still open
Long patterns	—	iterated limit $K_r^\infty/r^2 \Rightarrow \mathcal{W}$ with a Wasserstein bound (Theorem 18.1)

Table 3: Part II against Part I.

Part I	Part II at $r = 2$
Lemmas 2.2 and 2.3	Lemma 16.2: the same lemmas, with the same proof; they do not depend on r
Lemma 2.4, (2.4)	Lemma 16.3 at $r = 2$, proved through generating functions instead of the last reset
Theorem 3.1, (1.4)	Theorem 16.1 at $r = 2$: the same first-failure decomposition (the [write] note after its proof)
(1.5)	(16.11) at $r = 2$ (the [write] note after it)
Theorem 10.1	Theorem 17.2 at $r = 2$, since $L_2(y) = P(y)$; the same square-root transfer
(9.2), (9.6)	the same computation of $F_2(x, 2)$ (Section 21)
Theorem 10.2, (10.6)	the formulas (21.1), (21.2) and (21.3) at $r = 2$ (a [write] observation; Theorem 21.1 itself is proved for $r \geq 3$ only)

Table 4: Results of Part I proved again by manuscript 07 at $r = 2$.

Notion	Part II	Manuscript 07 (and the clash)
Chebyshev polynomial $\sum_j (-u)^j \binom{r+1-j}{j}$	$\Delta_r(u)$	$D_r(u)$ (Part I's descendant series $D_d(x)$ and deficit D_n)
descendant series $C(x)^{d+1}$	$D_d(x)$, as in Part I	H_d (Part I's polynomial factor $H_k(n)$)
coefficients of Δ_r	δ_j	d_j (its own constant d_r)
$-\rho_r \Delta'_r(\rho_r)$	ϑ_r	d_r (Part I's suffix state d and $d_1(k)$)
constants of the uniform estimate	Θ_r, Θ'_r	A_r, A'_r (its own safe-state series A_d at $d = r$; Part I's $A_k^*, A_{n,k}$)
prefix-deficit generating function	$\mathcal{G}_r(u, t), g_m(u)$	$G_r(z, t), g_m(z)$ (its geometric variables G_j ; Part I's G_k)
kernel weights $\rho_r^k 4^{-m} T_r(k+m, k)$	$\varpi_{k,m}$	$w_{k,m}$ (the exponent $w_n(s)$ of Lemma 22.1)
$\sqrt{1-4x}$ and $\sqrt{1-t}$	z , as in Part I	s (its window and Laplace variable s)
$\sqrt{1-y}$ in (18.4)	v	v (its $v = y_c/y$ of Section 21)
Bernoulli summand	I	B (its ballot numbers $B(m, p)$)
long-pattern limit variable	$\mathcal{W}$	W (Part I's Lambert W_{-1})
Wasserstein distance (cost $ x - y $)	d_W	W_1 (tempting misreading: Lambert's W_{-1})
window functions	$\varphi_n(s), \varphi(s)$	$f_n(s), f(s)$ (Part I's free energy $f(y)$)
order of an integer moment	h	ℓ (Part I's deficit $\ell = n - k$)
Fibonacci numbers	$\text{Fib}(j)$	F_j (its generating function F_r)

Table 5: Symbols of manuscript 07 renamed in Part II.

The following symbols keep manuscript 07's meaning inside Part II and mean something else in Part I. The unnormalized critical masses $p_{r,m}$ and their total P_r are not Part I's limit law p_k and its generating function $P(y)$; but $L_2(y) = P(y)$ (the [write] note after Theorem 17.2), and $p_{2,m} = b_m/2$ with b_m of (9.8) (the [write] note in Section 19). The deficit is M (Part I writes D_n or ℓ), and $M_r, M_{n,r}^{(y)}$ are not Part I's window normalization $M(\lambda)$. The ballot numbers are $B(m, p) = [x^m]C(x)^p$, so $B(m, j+1) = b(m, j)$ of (4.1). The root ρ_r of Δ_r lies in the u -plane; Part I's $\rho = 1/(2y)$ in the proof of Theorem 10.2 lies in the x -plane. The series $L_r(y)$ is not Part I's logarithmic series L_C, L_B ; a_r is not Part I's $a = \log 4$ or $a_m(k)$; $y_n(s)$ is not Part I's $y_n = 2e^{\lambda/n}$; $b_n = \frac{1}{2} \log n + \log \log n$ is not Part I's b_ℓ or $b = \frac{3}{2}$; the free energy f_r is Part I's f at $r = 2$. The geometric parameters q_j, q_0 , the number of roots h_r , the variables $\beta_y, \gamma_j, E_j, U_{r+1}$ (Chebyshev), η_r, ϵ_n , the generic p_m, P, a, α, L of Lemma 22.1, and the local $N = r + 2, S_r, Q_n, R$ are the manuscript's and unrelated to Part I's $q, h_0, h_1, \beta, \gamma_j, E_j, U, \eta_y, \epsilon, p_k, P, a, N, S(n, k), Q_k$ and $R_{n,k}$. The zeros $z_{n,j}$ of Theorem 23.1 are not values of $z = \sqrt{1-4x}$. The two figures are the delivered files and print the manuscript's legends: there $\pi_m = p_{3,m}/P_3$ is the limiting deficit law at $r = 3$, and a, P stand for a_3, P_3 .

15 The question and the contribution

An occurrence of a pattern in a permutation means an order-isomorphic subsequence; the positions need not be consecutive. For $r \geq 2$, put

$$\tau_r = 1(r+1)r \cdots 2.$$

For $\pi \in \text{Av}_n(123)$, let $K_r(\pi)$ be the largest k such that the subsequence consisting of the k largest values of π , in their original order, avoids τ_r . Thus $K_r = n$ when the whole permutation avoids both patterns. If $K_r = k < n$, the next smaller value is the first one whose inclusion creates τ_r .

Write

$$T_r(n, k) = \#\{\pi \in \text{Av}_n(123) : K_r(\pi) = k\}, \quad Z_{n,r}(y) = \sum_{k=1}^n T_r(n, k) y^k. \quad (15.1)$$

For real $y > 0$, the associated probability measure assigns weight $y^{K_r(\pi)}/Z_{n,r}(y)$ to π . We denote its prefix and deficit by $K_{n,r}^{(y)}$ and $M_{n,r}^{(y)} = n - K_{n,r}^{(y)}$. The word “prefix” refers to the successive restriction to largest values, not to consecutive positions of the original permutation.

At $r = 2$, T_r is David Callan’s triangle A273821 [1]. Its OEIS page still marks a generating function conjectural in the retrieved record (retrieved by the manuscript on 4 October 2026, local date). Part I of this report, however, already proves that formula and studies its critical beta law. Its Section 13, first proposed research question, specifically asks for the extension to $1(r+1)r \cdots 2$, including exact enumeration and critical laws. That is the question answered here. The inspected repository snapshot is 0f93381e8932d3aee3d5c2abfb5c98d128d49b9c. The source is in

SetTheory/Cardinals/docs/reports/enumerative-combinatorics/
a273821-first-pattern-failure/article.tex.

15.1 What is prior, and what is proved here

The unrefined enumeration of permutations avoiding both patterns is classical. Reverse-complement fixes 123 and changes τ_r to $r(r-1) \cdots 1(r+1)$. Chow–West [10], Theorem 3.1, give the corresponding Chebyshev quotient and growth constant; Krattenthaler [11], Theorem 9, supplies a Dyck-path formulation, while his Theorem 8 marks occurrences rather than first failure. Related stopping statistics in Catalan permutations also predate this work [12]. We rederive the needed safe-state formulas because the suffix information, rather than only the total count, is essential.

Statement	Contribution boundary
Safe-class Chebyshev quotient and growth	Classical Chow–West enumeration.
First-failure formula and beta law for $r = 2$	Already proved in Part I (the ProveIt report).
All- r first-failure refinement	Exact formula and finite positive coefficient sum proved here.
Uniform two-index estimate	Proved here; supplies the error control for critical limits and moments.
Critical laws for $r \geq 3$	Discrete heavy-tailed deficit, sharp variation rate, moment threshold.
Shifted coexistence and local complex zeros	Proved here with uniform complex error control.
Growing-pattern behavior	Iterated limit and explicit Wasserstein bound; no joint $r = r(n)$ theorem.

Table 6: The precise scope of the article. The literature search is targeted; this ledger does not assert that no equivalent refinement exists elsewhere.

15.2 A preview of the change of critical behavior

Define

$$\rho_r = \frac{1}{4 \cos^2(\pi/(r+2))}, \quad \Lambda_r = \rho_r^{-1}, \quad y_c = 4\rho_r. \quad (15.2)$$

For $r = 2$, $y_c = 2$; for $r \geq 3$, $1 < y_c < 2$. The following comparison concerns fixed r and $n \rightarrow \infty$.

Weight	$r = 2$, prior result	$r \geq 3$, present result
$0 < y < y_c$	K has a discrete limit	K has an explicit geometric-sum limit
$y = y_c$	$K/n \Rightarrow \text{Beta}(1, 1/2)$	$n - K$ has a discrete limit; $K/n \rightarrow 1$
$y > y_c$	$n - K$ has a discrete limit	$n - K$ has an exponentially decaying limit
Coexistence	A window of order $1/n$ at 2	A window of order $1/n$, displaced by $(\log n)/n$

Table 7: The smallest pattern is exceptional. The critical deficit for $r \geq 3$ is tight but its finite-size mean diverges as $\sqrt{n}$.

The asymptotic estimates below use conventional singularity analysis [13]. The key uniform estimate is proved directly by one-variable coefficient extraction and a contour bound, so the moving-window argument does not rely on an unproved interchange of limits.

16 Exact first-failure enumeration

Put

$$C(x) = \frac{2}{1 + \sqrt{1 - 4x}}, \quad \Delta_r(u) = \sum_{j=0}^{\lfloor (r+1)/2 \rfloor} (-u)^j \binom{r+1-j}{j}. \quad (16.1)$$

The square root is the germ equal to one at the origin. In particular, $C(0) = 1$, $C = 1 + xC^2$, and

$$\Delta_0 = 1, \quad \Delta_1 = 1 - u, \quad \Delta_r = \Delta_{r-1} - u\Delta_{r-2}.$$

Theorem 16.1 (The whole first-failure family). *For every integer $r \geq 2$, as an identity of formal power series,*

$$F_r(x, y) := \sum_{n \geq 1} \sum_{k=1}^n T_r(n, k) x^n y^k = \frac{u\Delta_{r-2}(u)}{\Delta_r(u)} + \frac{xu^r C(x)^{r+1}}{\Delta_r(u)(1 - uC(x))}, \quad u = xy. \quad (16.2)$$

Let $e_h = [u^h]\Delta_r(u)^{-1}$, with $e_h = 0$ for $h < 0$, and set

$$B(m, p) = [x^m]C(x)^p = \frac{p}{2m+p} \binom{2m+p}{m} \quad (m \geq 0, p \geq 1). \quad (16.3)$$

Then, for $n > k \geq r$,

$$T_r(n, k) = \sum_{j=r}^k e_{k-j} B(n-k-1, j+1). \quad (16.4)$$

Off the diagonal, $T_r(n, k) = 0$ when $k < r$. On the diagonal,

$$\sum_{n \geq 1} T_r(n, n) u^n = \frac{u\Delta_{r-2}(u)}{\Delta_r(u)}, \quad 1 + \sum_{n \geq 1} T_r(n, n) u^n = \frac{\Delta_{r-1}(u)}{\Delta_r(u)}. \quad (16.5)$$

All terms in (16.4) are nonnegative. Every row sums to the Catalan number $C_n = (n+1)^{-1} \binom{2n}{n}$.

16.1 Insertion of successive minima

Deleting the minimum of a nonempty permutation and standardizing gives its unique parent. Reverse this operation, starting from the one-entry permutation. Let d be the length of the terminal decreasing run of consecutive positions. An insertion site is indexed by the number j of old entries to its right.

Lemma 16.2 (Active sites and descendants). *In a 123-avoiding permutation of state d , insertion of a new minimum preserves 123 avoidance precisely at $0 \leq j \leq d$. Appending it, $j = 0$, gives state $d + 1$; the other sites give state j . The generating function for descendants obtained by any number of further allowed insertions is $C(x)^{d+1}$.*

Proof. Any new increasing triple must start at the inserted minimum. Such a triple exists exactly when the suffix to its right contains an increasing pair. That suffix must therefore decrease, proving the active-site rule. If the minimum is not appended, the decreasing suffix on its right is separated from the preceding entries by an ascent, proving the state rule.

For the descendant series D_d , this gives

$$D_d = 1 + x \sum_{j=1}^{d+1} D_j.$$

The solution with all constant terms one is unique coefficient by coefficient: the coefficient of degree m uses only degree $m - 1$ at finitely many states. The proposed solution satisfies it because

$$x \sum_{j=1}^{d+1} C^{j+1} = (C - 1)(1 + C + \cdots + C^d) = C^{d+1} - 1.$$

Starting from state one, the number of size- n permutations is $[x^{n-1}]C^2 = C_n$. □

[write] **Added 5 October 2026, batch 98: a second route.** Lemma 16.2 is Part I's Lemma 2.2 together with Lemma 2.3, proved in the same way; it does not involve r . The manuscript names the descendant series H_d ; it is Part I's D_d .

If the current permutation avoids τ_r , a new occurrence has to start at the new minimum. At an active site the right suffix decreases, so the insertion creates τ_r exactly when $j \geq r$. Safe insertions thus have the state rule

$$d \longrightarrow d + 1, 1, 2, \dots, \min(d, r - 1). \quad (16.6)$$

An occurrence, once created, cannot disappear when smaller entries are inserted. This monotonicity makes a decomposition at the first failing insertion exact.

16.2 The safe suffix distribution

Let $a_{k,d}$ count safe size- k permutations of suffix state d , and $A_d(u) = \sum_{k \geq 1} a_{k,d} u^k$.

Lemma 16.3 (Safe states and their tails). *For $1 \leq d \leq r - 1$,*

$$A_d(u) = \frac{u^d \Delta_{r-d}(u)}{\Delta_r(u)}, \quad (16.7)$$

and, for $d \geq r - 1$,

$$A_d(u) = \frac{u^d (1 - u)}{\Delta_r(u)}.$$

The two expressions agree at $d = r - 1$. In particular,

$$\sum_{d \geq 1} A_d(u) = \frac{u\Delta_{r-2}(u)}{\Delta_r(u)}, \quad \sum_{d \geq j} A_d(u) = \frac{u^j}{\Delta_r(u)} \quad (j \geq r). \quad (16.8)$$

Proof. With $A_0 = 0$, the state equations from (16.6) are

$$A_d = \mathbf{1}_{d=1}u + uA_{d-1} + u \sum_{e \geq d} A_e \quad (1 \leq d \leq r-1), \quad A_d = uA_{d-1} \quad (d \geq r).$$

They determine every coefficient uniquely; in size k only states $d \leq k$ occur. For the proposed expressions, summation gives

$$\sum_{e \geq d} A_e = \frac{u^d \Delta_{r-d-1}(u)}{\Delta_r(u)} \quad (1 \leq d \leq r-1).$$

This identity follows by telescoping the recurrence for Δ , with the geometric tail at $d = r - 1$. For $d \geq 2$, subtraction of the two successive formulas for A_d then gives the required equation. At $d = 1$, it is the identity $\Delta_{r-1} = \Delta_r + u\Delta_{r-2}$. Summing the geometric tail proves the second assertion of (16.8). $\square$

[write] Added 5 October 2026, batch 98: the case $r = 2$. At $r = 2$, $\Delta_2 = 1 - 2u$ and every state is covered by the second formula: $A_d(u) = u^d(1 - u)/(1 - 2u)$, whose coefficients are $a_{d,d} = 1$ and $a_{k,d} = 2^{k-d-1}$ for $k > d$; the tail is $u^j/(1 - 2u)$. This is Part I's Lemma 2.4 with (2.4), obtained from the state equations instead of the position of the last reset.

Proof of Theorem 16.1. The diagonal is the safe class and contributes the first term of (16.2). Suppose the first failure occurs after k safe entries, at a site with $j \geq r$ entries to the right. The safe prefix must have suffix state at least j ; its generating function is $u^j/\Delta_r(u)$. The failing insertion contributes x , not xy , because it does not increase K_r . By Lemma 16.2, subsequent insertions contribute $C(x)^{j+1}$. Summing gives

$$\frac{x}{\Delta_r(u)} \sum_{j \geq r} u^j C(x)^{j+1} = \frac{xu^r C(x)^{r+1}}{\Delta_r(u)(1 - uC(x))}.$$

Coefficient extraction proves (16.4); the ballot identity (16.3) follows from Lagrange inversion of $C = 1 + xC^2$. Positivity of e_h also follows from the positive-root factorization in Lemma 17.1 below. The insertion decomposition partitions all 123-avoiding permutations, giving the row sums. $\square$

[write] Added 5 October 2026, batch 98: a second route. At $r = 2$, $e_h = 2^h$ and $B(m, j+1) = b(m, j)$ of (4.1), so (16.4) is Part I's convolution (3.1), the diagonal (16.5) is $u/(1 - 2u)$, and (16.2) is (1.4). The proof above is the proof of Theorem 3.1 carried out for every r . The diagonal for general r is the classical Chow–West safe count [10].

16.3 Specializations and an exact recurrence

For $r = 2$, $\Delta_2 = 1 - 2u$, and (16.2) becomes exactly the repository's proved A273821 formula (1.4). For $r = 3$,

$$F_3(x, y) = \frac{u(1 - u)}{1 - 3u + u^2} + \frac{xu^3 C(x)^4}{(1 - 3u + u^2)(1 - uC(x))}. \quad (16.9)$$

Its diagonal is $T_3(n, n) = \text{Fib}(2n - 1)$, with the usual Fibonacci convention $\text{Fib}(1) = \text{Fib}(2) = 1$. This diagonal identity is classical.

[write] Added 5 October 2026, batch 98: a proof of the diagonal identity. The manuscript cites no source for it; it follows from (16.9). Since $\text{Fib}(m+2) = 3\text{Fib}(m) - \text{Fib}(m-2)$, the sequence $\psi_n = \text{Fib}(2n-1)$ satisfies $\psi_n = 3\psi_{n-1} - \psi_{n-2}$ for $n \geq 3$, with $\psi_1 = 1, \psi_2 = 2$. Hence $(1 - 3u + u^2) \sum_{n \geq 1} \psi_n u^n = u - u^2$, which is the diagonal term $u(1-u)/(1-3u+u^2)$ of (16.9). (Checked by series expansion through $n = 40$ in this write.)

Let $\delta_j = (-1)^j \binom{r+1-j}{j}$. Multiplication by Δ_r gives, off the diagonal and for $n > k \geq r$,

$$T_r(n, k) = - \sum_{j=1}^{\lfloor (r+1)/2 \rfloor} \delta_j T_r(n-j, k-j) + B(n-k-1, k+1). \quad (16.10)$$

Terms outside the defined triangle are zero. The diagonal is supplied by (16.5). This recurrence fills all rows through size N in $O(rN^2)$ arithmetic operations, apart from the bit cost of the integer arithmetic and ballot coefficients.

For the first possible failing column,

$$T_r(n, r) = B(n-r-1, r+1) \quad (n > r).$$

More generally, every fixed column $k \geq r$ has growth constant four:

$$T_r(n, k) \sim \frac{4^n}{\sqrt{\pi} n^{3/2}} \sum_{j=r}^k e_{k-j} \frac{(j+1)2^j}{4^{k+1}}. \quad (16.11)$$

This follows term by term from the finite positive formula and Stirling's formula for (16.3). Thus the fixed-column part of the repository question has a positive explicit answer for every r .

[write] Added 5 October 2026, batch 98: the case $r = 2$. With $e_h = 2^h$, the constant in (16.11) is $2^{-k-2} \sum_{j=2}^k (j+1) = (k-1)(k+4)/2^{k+3} = p_k$, so (16.11) at $r = 2$ is Part I's (1.5); Part I's Theorem 5.2 gives the full expansion there.

17 The spectral data and the subcritical law

Lemma 17.1 (Classical Chebyshev spectrum). *The polynomial Δ_r satisfies*

$$\Delta_r(u) = u^{(r+1)/2} U_{r+1}\left(\frac{1}{2\sqrt{u}}\right), \quad (17.1)$$

understood as a polynomial identity. Its simple roots are

$$\rho_{r,j} = \frac{1}{4 \cos^2(j\pi/(r+2))}, \quad 1 \leq j \leq h_r := \lfloor (r+1)/2 \rfloor.$$

They are positive and increasing. In particular, $\rho_r = \rho_{r,1}$ is the unique root of smallest modulus and

$$\vartheta_r := -\rho_r \Delta'_r(\rho_r) > 0. \quad (17.2)$$

Proof. The right side of (17.1) has the same recurrence and first two values as Δ_r . The usual formula $U_j(\cos \theta) = \sin((j+1)\theta)/\sin \theta$ gives the roots, with the opposite cosine roots paired by squaring. Consequently $\Delta_r(u) = \prod_{j=1}^{h_r} (1 - u/\rho_{r,j})$. The derivative at the first root is negative. This also proves the nonnegativity of the coefficients of $1/\Delta_r$. $\square$

Define, for $0 < y < y_c$,

$$L_r(y) = \frac{2^{r-2}(y/4)^r((r+1) - ry/2)}{\Delta_r(y/4)(1 - y/2)^2}. \quad (17.3)$$

In particular $L_r(1) = 1$, as follows either from Catalan row sums or from $\Delta_r(1/4) = (r+2)/2^{r+1}$.

Theorem 17.2 (Subcritical prefix distribution). *For fixed $r \geq 2$ and $0 < y < y_c$,*

$$Z_{n,r}(y) = \frac{4^n}{\sqrt{\pi} n^{3/2}} (L_r(y) + O(n^{-1})). \quad (17.4)$$

The random variable $K_{n,r}^{(y)}$ converges to a discrete variable $K_r^{(y)}$ with probability generating function

$$\mathbb{E}_t K_r^{(y)} = \frac{L_r(yt)}{L_r(y)}. \quad (17.5)$$

For some $R > 1$,

$$\sum_{k \geq 0} R^k \left| \mathbb{P}(K_{n,r}^{(y)} = k) - \mathbb{P}(K_r^{(y)} = k) \right| = O(n^{-1}).$$

The constants may depend on r and y . In particular, all fixed moments converge.

Proof. Let $z = \sqrt{1 - 4x}$. In (16.2), $u = xy = y(1 - z^2)/4$ has no term linear in z , while $C(x) = 2 - 2z + O(z^2)$. Direct expansion gives the coefficient of z as $-2L_r(y)$. Both denominators stay away from zero at $x = 1/4$ for $|y| < y_c$, because $y_c \leq 2$. The expression is analytic in a dented neighborhood of that point, uniformly when y stays in a smaller disk. Expansion through the next odd power and the coefficient formula for $(1 - 4x)^{1/2}$ prove (17.4). The remaining integer powers are analytic locally and the next singular contribution has order $4^n n^{-5/2}$. This is the elementary square-root case of the transfer principle [13].

Apply the same expansion at yt on a fixed circle $|t| = R_1 > 1$ with $yR_1 < y_c$, and divide by (17.4). The resulting probability generating functions differ uniformly by $O(n^{-1})$. Cauchy's coefficient estimate gives an error $O(R_1^{-k} n^{-1})$. Summing after multiplication by any fixed $1 < R < R_1$ proves the assertion, including that the limiting coefficients form a probability distribution. $\square$

[write] Added 5 October 2026, batch 98: the omitted expansion, and the case $r = 2$. The manuscript states the coefficient of z without the computation. The first term of (16.2) is a function of u alone, hence even in z . In the second, $xu^r/\Delta_r(u)$ is even in z and equals $\frac{1}{4}(y/4)^r/\Delta_r(y/4)$ at $z = 0$, while $C = 2/(1 + z)$ gives $dC/dz = -2$ at $z = 0$, so

$$\frac{d}{dz} \frac{C^{r+1}}{1 - uC} \Big|_{z=0} = -2 \frac{(r+1)2^r(1 - y/2) + 2^{r+1}y/4}{(1 - y/2)^2} = -2^{r+1} \frac{(r+1) - ry/2}{(1 - y/2)^2}.$$

Multiplying, the coefficient of z is $-2L_r(y)$, and $[x^n](1 - 4x)^{1/2} = -\frac{4^n}{2\sqrt{\pi}n^{3/2}}(1 + O(n^{-1}))$ gives the leading term of (17.4). (Checked symbolically for $r = 2, \dots, 7$ in this write.)

At $r = 2$, $\Delta_2(y/4) = 1 - y/2$ and $L_2(y) = y^2(3 - y)/(2(2 - y)^3) = P(y)$ of (6.1). Since $C_n = \frac{4^n}{\sqrt{\pi}n^{3/2}}(1 + O(n^{-1}))$, Theorem 17.2 at $r = 2$ is Part I's Theorem 10.1 with the weighted error of Theorem 6.1, proved by the same square-root transfer.

17.1 A positive independent-sum representation

Our convention is that $G \sim \text{Geom}_0(q)$, for $0 \leq q < 1$, means $\mathbb{P}(G = m) = (1 - q)q^m$ for $m \geq 0$.

Theorem 17.3 (Geometric decomposition). *Put*

$$q_j = y \cos^2 \frac{j\pi}{r+2} \quad (1 \leq j \leq h_r), \quad q_0 = y/2, \quad \beta_y = \frac{1}{1+r(1-q_0)}.$$

For $0 < y < y_c$, all $q_j, q_0 < 1$, and

$$K_r^{(y)} \stackrel{d}{=} r + \sum_{j=1}^{h_r} G_j + G + IG', \quad (17.6)$$

where all variables are independent, $G_j \sim \text{Geom}_0(q_j)$, $G, G' \sim \text{Geom}_0(q_0)$, and $I \sim \text{Bernoulli}(\beta_y)$.

Proof. Factor $\Delta_r(yt/4)$ using Lemma 17.1. After extracting its normalized geometric factors and t^r , the remaining factor of $L_r(yt)/L_r(y)$ is

$$\frac{1-q_0}{1-q_0t} \left(1 - \beta_y + \beta_y \frac{1-q_0}{1-q_0t} \right).$$

This is the generating function of $G + IG'$, proving the claim. $\square$

At $y = 1$, write $K_r^\infty = K_r^{(1)}$. The distribution of $V_r = G + IG'$ becomes especially simple:

$$\mathbb{P}(V_r = m) = \frac{r+m+1}{(r+2)2^{m+1}}, \quad m \geq 0. \quad (17.7)$$

Logarithmic differentiation of Δ_r , or the independent sum, gives

$$\mathbb{E}K_r^\infty = \frac{r(r+1)}{6} + r + 1 + \frac{2}{r+2}, \quad (17.8)$$

$$\text{Var } K_r^\infty = \frac{r(r+1)(r+3)(r+4)}{90} + 2 + \frac{6}{r+2} - \frac{4}{(r+2)^2}. \quad (17.9)$$

For example, $\mathbb{E}K_3^\infty = 32/5$. The identities needed for these computations are

$$\Delta_r(1/4) = \frac{r+2}{2^{r+1}}, \quad -\frac{\Delta'_r(1/4)}{4\Delta_r(1/4)} = \frac{r(r+1)}{6},$$

and the corresponding second logarithmic derivative. They follow by expanding the two-root formula for Δ_r in Section 18.

[write] Added 5 October 2026, batch 98: the case $r = 2$. At $r = 2$, $h_2 = 1$ and $q_1 = y/2 = q_0$: the limit law of Part I's Theorem 10.1 is that of $2 + G_1 + G + IG'$ with three independent $\text{Geom}_0(y/2)$ variables and $I \sim \text{Bernoulli}(1/(3-y))$. At $y = 1$ this gives mean $9/2$ and variance $21/4$, as in (6.1). Part I does not contain this representation.

For fixed $r \geq 3$, as $y \uparrow y_c$, only $q_1 = y/y_c$ approaches one. All the other independent summands stay tight. Therefore

$$(1 - y/y_c)K_r^{(y)} \Rightarrow \text{Exp}(1). \quad (17.10)$$

For $r = 2$, all three geometric contributions become large and $\beta_y \rightarrow 1$, giving instead a gamma limit of shape three in this *iterated* limit. Neither assertion substitutes for a theorem with $y = y(n)$; that moving problem is handled in Section 22.

18 An iterated limit for long forbidden patterns

The geometric decomposition makes it possible to let the pattern length grow *after* the permutation-size limit has been taken. It also gives a quantitative comparison rather than only a Laplace-transform calculation.

Theorem 18.1 (Long-pattern law with an explicit rate). *Let E_j , $j \geq 1$, be independent exponentials of rate one and put*

$$\mathcal{W} = \sum_{j \geq 1} \frac{E_j}{\pi^2 j^2}. \quad (18.1)$$

The sum is finite almost surely and in L^1 . With d_W denoting the Wasserstein distance for the cost $|x - y|$,

$$d_W\left(\mathcal{L}\left(\frac{K_r^\infty}{(r+2)^2}\right), \mathcal{L}(\mathcal{W})\right) \leq \frac{9r^2 + 28r + 32}{6(r+2)^3}. \quad (18.2)$$

In particular, $K_r^\infty / r^2 \Rightarrow \mathcal{W}$, and

$$\mathbb{E}e^{-s\mathcal{W}} = \frac{\sqrt{s}}{\sinh \sqrt{s}} \quad (s \geq 0), \quad \mathbb{E}\mathcal{W} = \frac{1}{6}, \quad \text{Var } \mathcal{W} = \frac{1}{90}. \quad (18.3)$$

The quotient at $s = 0$ is interpreted by continuity.

Proof. Let $N = r + 2$, $h = h_r$, and $\gamma_j = -2 \log \cos(\pi j / N)$. Couple the geometricals in (17.6), at $y = 1$, by

$$G_j = \lfloor E_j / \gamma_j \rfloor.$$

Indeed $\mathbb{P}(G_j \geq m) = e^{-m\gamma_j} = \cos^{2m}(\pi j / N)$. Since $-2 \log \cos \theta \geq \theta^2$ on $[0, \pi/2)$,

$$0 \leq S_r := \frac{1}{N^2} \sum_{j=1}^h G_j \leq \mathcal{W}.$$

The trigonometric sum implicit in (17.8) gives

$$\mathbb{E}(\mathcal{W} - S_r) = \frac{1}{6} - \frac{r(r+1)}{6N^2} = \frac{3r+4}{6N^2}.$$

Choose V_r independently, as in (17.7); then $\mathbb{E}V_r = 1 + 2/N$. The triangle inequality in this coupling yields

$$\mathbb{E} \left| \frac{r + V_r}{N^2} + S_r - \mathcal{W} \right| \leq \frac{r + 1 + 2/N}{N^2} + \frac{3r + 4}{6N^2},$$

which simplifies to (18.2). The series for $\mathcal{W}$ has finite expectation $\sum_j (\pi^2 j^2)^{-1} = 1/6$, hence is finite almost surely. Independence and monotone convergence give the product of its Laplace transforms. Euler's product for $\sinh$, recorded in [14], reduces that product to (18.3). Summing the variances gives $1/90$. Finally $N^2/r^2 \rightarrow 1$. $\square$

For a direct algebraic check, put $v = \sqrt{1 - y}$. The recurrence for Δ_r gives

$$\Delta_r(y/4) = \frac{(1+v)^{r+2} - (1-v)^{r+2}}{2^{r+2}v}. \quad (18.4)$$

At $y = e^{-s/r^2}$, $rv \rightarrow \sqrt{s}$; substitution into (17.3) again gives $L_r(e^{-s/r^2}) \rightarrow \sqrt{s}/\sinh \sqrt{s}$. This is an independent consistency check of the scale and constants.

The order of limits matters. The theorem concerns $\lim_{r \rightarrow \infty} \lim_{n \rightarrow \infty}$, with r fixed in the inner limit. No assertion is made here for arbitrary choices $r = r(n)$. The spectral gap used later shrinks with r , so such a joint theorem requires additional estimates.

19 A uniform estimate in prefix size and deficit

From this point through Section 23, fix $r \geq 3$. We often omit r from ρ_r, ϑ_r and from constants. The strict inequality $\rho < 1/2$ is essential.

Define a series with nonnegative coefficients by

$$\mathcal{P}_r(t) = \frac{1}{\vartheta_r} \left[\rho_r \Delta_{r-2}(\rho_r) + \frac{(t/4)\rho_r^r C(t/4)^{r+1}}{1 - \rho_r C(t/4)} \right] = \sum_{m \geq 0} p_{r,m} t^m. \quad (19.1)$$

It has finite positive total mass

$$P_r = \mathcal{P}_r(1) = \frac{1}{\vartheta_r} \left[\rho_r \Delta_{r-2}(\rho_r) + \frac{2^{r-1} \rho_r^r}{1 - 2\rho_r} \right]. \quad (19.2)$$

Set

$$a_r = \frac{2^{r-2} \rho_r^r ((r+1) - 2r\rho_r)}{\vartheta_r \sqrt{\pi} (1 - 2\rho_r)^2} > 0. \quad (19.3)$$

These are unnormalized coefficients: the probability law used below will have probabilities $p_{r,m}/P_r$.

Lemma 19.1 (The residue tail). *As $m \rightarrow \infty$,*

$$p_{r,m} = a_r m^{-3/2} + O(m^{-5/2}). \quad (19.4)$$

Consequently

$$\sum_{m > n} p_{r,m} = 2a_r n^{-1/2} + O(n^{-3/2}). \quad (19.5)$$

Proof. With $z = \sqrt{1-t}$, the expansion $C(t/4) = 2 - 2z + 2z^2 + O(z^3)$ gives

$$\mathcal{P}_r(t) = P_r - 2\sqrt{\pi} a_r \sqrt{1-t} + O(1-t). \quad (19.6)$$

For clarity, the linear coefficient before division by ϑ_r is minus

$$\frac{2^{r-1} \rho_r^r}{1 - 2\rho_r} \left(r + 1 + \frac{2\rho_r}{1 - 2\rho_r} \right).$$

There is a convergent expansion in powers of z near zero. The next noninteger power of $1-t$ is $(1-t)^{3/2}$. On $|t| \leq 1$, $\rho_r |C(t/4)| \leq 2\rho_r < 1$, so the denominator does not vanish; the same holds in a sufficiently small dented extension near $t = 1$. There are no other singularities on the unit circle. Applying the square-root coefficient formula, including its next-order error, yields (19.4). Finally $\sum_{m > n} m^{-3/2} = 2n^{-1/2} + O(n^{-3/2})$, by integral comparison with the one-step endpoint error; the sum of the remainder is $O(n^{-3/2})$. $\square$

Theorem 19.2 (Uniform two-index residue approximation). *There exist constants $\Theta_r < \infty$ and $0 < \eta_r < 1$ such that*

$$\left| \rho_r^k 4^{-m} T_r(k+m, k) - p_{r,m} \right| \leq \Theta_r \eta_r^k (m+1)^{-3/2} \quad (k, m \geq 0). \quad (19.7)$$

Here $T_r(0, 0) = 0$, and all entries with $k = 0$ are zero. The constants depend on r , but neither index is restricted.

Proof. Mark prefix and deficit separately:

$$\mathcal{G}_r(u, t) := \sum_{k, m \geq 0} T_r(k + m, k) u^k t^m = \frac{u \Delta_{r-2}(u) + t u^r C(t)^{r+1} / (1 - u C(t))}{\Delta_r(u)}. \quad (19.8)$$

Choose $\rho_r < R < \min(\rho_{r,2}, 1/2)$, where a missing second root is omitted. On $|u| = R$, $\Delta_r(u)$ is bounded away from zero. For each fixed m , let $g_m(u) = [t^m] \mathcal{G}_r(u, t/4)$. The only pole of g_m between a small circle about zero and $|u| = R$ is the simple pole at $u = \rho_r$. Its principal part is $p_{r,m}/(1 - u/\rho_r)$. Thus Cauchy's formula and the residue theorem give

$$4^{-m} T_r(k + m, k) - p_{r,m} \rho_r^{-k} = \frac{1}{2\pi i} \int_{|u|=R} g_m(u) u^{-k-1} du.$$

It remains to bound g_m uniformly in m . For integers $p, m \geq 1$,

$$[t^m] C(t/4)^p = 4^{-m} \frac{p}{2m+p} \binom{2m+p}{m} \leq \frac{p 2^{p-1}}{\sqrt{\pi} m^{3/2}}. \quad (19.9)$$

Indeed $\binom{2m+p}{m} \leq 2^p \binom{2m}{m}$, since each successive ratio $(2m+j)/(m+j) \leq 2$, and $\binom{2m}{m} \leq 4^m / \sqrt{\pi m}$. Expand $(1 - uC)^{-1}$ geometrically in the numerator of (19.8). Bound (19.9) reduces the resulting majorant to a constant times $\sum_{j \geq 0} (r+j+1)(2R)^j$, which converges. The leading factor $t/4$ only shifts the coefficient index; the cases $m = 0, 1$ have finite bounds and can be absorbed into the constant. Therefore

$$\sup_{|u|=R} |g_m(u)| \leq \Theta'_r(m+1)^{-3/2}.$$

Multiply the contour estimate by ρ_r^k , and take $\eta_r = \rho_r/R < 1$. □

Remark 19.3 (Why this estimate is stronger than a pointwise limit). For each fixed deficit, a simple residue calculation already predicts $p_{r,m}$. It does not by itself control deficits growing with n , moments dominated by rare large deficits, or a moving weight. The summable factor $(m+1)^{-3/2}$ in (19.7) supplies exactly those missing controls. The possibility of choosing $R < 1/2$ is the analytic distinction between $r \geq 3$ and $r = 2$.

[write] **Added 5 October 2026, batch 98: the series at $r = 2$, and why the estimate fails there.** The definitions (19.1) make sense at $r = 2$: $\rho_2 = \frac{1}{2}$, $\Delta_0 = 1$, $\vartheta_2 = 1$, and, writing $z = \sqrt{1-t}$ so that $C(t/4) = 2/(1+z)$ and $t = (1-z)(1+z)$,

$$\mathcal{P}_2(t) = \frac{1}{2} + \frac{t}{16} \frac{C(t/4)^3}{1 - C(t/4)/2} = \frac{1}{2} + \frac{1-z}{2z(1+z)} = \frac{1}{2} + \frac{1}{2}((1-t)^{-1/2} - C(t/4)).$$

By the series identity in the proof of Theorem 10.2, $p_{2,0} = \frac{1}{2}$ and $p_{2,m} = 4^{-m} \binom{2m}{m-1}/2$ for $m \geq 1$: $p_{2,m} = b_m/2$ for the positive critical weights b_m of Part I's (9.8). (Checked through $m = 30$ in this write.) Their sum diverges, because $b_m \sim (\pi m)^{-1/2}$; this is the vanishing of $1 - 2\rho_2$ in (19.2), and it is why the critical law at $r = 2$ is Part I's beta law instead of Theorem 20.1. The pointwise limit $\rho_2^k 4^{-m} T_2(k+m, k) \rightarrow p_{2,m}$ as $k \rightarrow \infty$ still holds, by (8.7). But (19.7) is false at $r = 2$ for every choice of constants: since $T_2(1+m, 1) = 0$ for $m \geq 1$, its left side at $k = 1$ is $p_{2,m} \sim \frac{1}{2}(\pi m)^{-1/2}$, which is not $O((m+1)^{-3/2})$. For $r \geq 3$ the same column gives $p_{r,m} \sim a_r m^{-3/2}$, consistent with (19.7). So the restriction $r \geq 3$ of Sections 19–23 is necessary for this estimate, not only for its proof.

20 Critical condensation and the full moment threshold

Theorem 20.1 (Critical partition function and deficit law). *At $y = y_c$,*

$$Z_{n,r}(y_c) = 4^n \left(P_r - \frac{2a_r}{\sqrt{n}} + O_r(n^{-3/2}) \right). \quad (20.1)$$

The deficit $M_{n,r}^{(y_c)}$ converges to a variable M_r with

$$\mathbb{P}(M_r = m) = \frac{p_{r,m}}{P_r}, \quad \mathbb{E}t^{M_r} = \frac{\mathcal{P}_r(t)}{P_r}. \quad (20.2)$$

With total variation defined as one half the sum of absolute mass differences, the convergence rate has the sharp leading term

$$d_{\text{TV}}(\mathcal{L}(M_{n,r}^{(y_c)}), \mathcal{L}(M_r)) = \frac{2a_r}{P_r \sqrt{n}} + O_r(n^{-3/2}). \quad (20.3)$$

Moreover,

$$\mathbb{P}(M_r = m) \sim \frac{a_r}{P_r} m^{-3/2}, \quad \mathbb{P}(M_r \geq m) \sim \frac{2a_r}{P_r} m^{-1/2}. \quad (20.4)$$

In particular, $K_{n,r}^{(y_c)}/n \rightarrow 1$ in probability.

Proof. Write $\varpi_{k,m} = \rho_r^k 4^{-m} T_r(k+m, k)$. At $y_c = 4\rho_r$,

$$4^{-n} Z_{n,r}(y_c) = \sum_{m=0}^n \varpi_{n-m,m}.$$

For any fixed $0 < \eta < 1$, splitting at $k = n/2$ shows

$$\sum_{k=0}^n \eta^k (n-k+1)^{-3/2} = O(n^{-3/2}). \quad (20.5)$$

Thus Theorem 19.2 replaces the sum by $\sum_{m \leq n} p_{r,m}$ with error $O(n^{-3/2})$. Equation (19.5) proves (20.1).

Let Q_n be the law with probabilities $p_{r,m}/\sum_{j \leq n} p_{r,j}$, for $0 \leq m \leq n$. The sum of absolute errors in the unnormalized weights is $O(n^{-3/2})$, so normalization gives $d_{\text{TV}}(\mathcal{L}(M_{n,r}^{(y_c)}), Q_n) = O(n^{-3/2})$. The total variation between a probability law and its conditioning to $m \leq n$ is exactly its omitted tail mass. Hence

$$d_{\text{TV}}(Q_n, \mathcal{L}(M_r)) = \frac{1}{P_r} \sum_{m > n} p_{r,m}.$$

The triangle inequality, in both directions, now proves (20.3). The tail statements follow from Lemma 19.1. Tightness of the deficits implies $M_{n,r}^{(y_c)}/n \rightarrow 0$. $\square$

The limiting deficit has infinite mean. This is compatible with tightness: most probability mass remains near small deficits, while rare large deficits dominate the expectation. The following theorem quantifies this distinction for every positive moment.

Theorem 20.2 (Critical moment threshold). *For real $b > 0$, as $n \rightarrow \infty$,*

$$\mathbb{E}(M_{n,r}^{(y_c)})^b = \begin{cases} \mathbb{E}M_r^b + o(1), & 0 < b < 1/2, \\ (a_r/P_r) \log n + O_r(1), & b = 1/2, \\ \frac{a_r}{P_r(b-1/2)} n^{b-1/2} (1 + o(1)), & b > 1/2. \end{cases} \quad (20.6)$$

In particular,

$$\mathbb{E}M_{n,r}^{(y_c)} \sim \frac{2a_r}{P_r} \sqrt{n}, \quad \text{Var}(M_{n,r}^{(y_c)}) \sim \frac{2a_r}{3P_r} n^{3/2}. \quad (20.7)$$

For every positive integer h , both ordinary and falling-factorial moments have leading term

$$\mathbb{E}(M_{n,r}^{(y_c)})^h \sim \mathbb{E}(M_{n,r}^{(y_c)})_h \sim \frac{2a_r}{P_r(2h-1)} n^{h-1/2}. \quad (20.8)$$

Proof. After multiplying (19.7) by m^b and summing along $k + m = n$, the error is $O(n^{b-3/2})$: for $k \leq n/2$ the factor $m^b(m+1)^{-3/2}$ is $O(n^{b-3/2})$, and for $k > n/2$ exponential decay absorbs any fixed power of n . Thus the unnormalized moment is governed by $\sum_{m \leq n} m^b p_{r,m}$. Insert (19.4). The leading power sum converges for $b < 1/2$, is logarithmic at $b = 1/2$, and has equivalent $n^{b-1/2}/(b-1/2)$ for $b > 1/2$. The next-order tail gives a smaller contribution in every case. Division by (20.1) proves (20.6). For the variance, the squared mean is $O(n)$, lower order than $n^{3/2}$. A falling factorial is a monic polynomial whose other terms have smaller degree, proving (20.8). $\square$

21 Supercritical weights and the phase diagram

Theorem 21.1 (Supercritical deficit as an exponential tilt). *For fixed $y > y_c$, let $v = y_c/y < 1$. Then, for some $0 < \xi < 1$,*

$$Z_{n,r}(y) = \left(\frac{y}{\rho_r}\right)^n (\mathcal{P}_r(v) + O_r(\xi^n)). \quad (21.1)$$

The limiting deficit law is

$$\mathbb{P}(M_r^{(y)} = m) = \frac{p_{r,m} v^m}{\mathcal{P}_r(v)}, \quad \mathbb{E}_t M_r^{(y)} = \frac{\mathcal{P}_r(vt)}{\mathcal{P}_r(v)}. \quad (21.2)$$

Convergence in total variation is exponentially fast for each fixed $y > y_c$. The constants and ξ may depend on y .

Proof. Normalize (15.1) by $(y/\rho_r)^n$ and put $m = n - k$. The sum becomes $\sum_{m \leq n} v^m \varpi_{n-m,m}$. The uniform estimate replaces this by $\sum_{m \leq n} v^m p_{r,m}$, with absolute error bounded by a constant times $\sum_{m \leq n} v^m \eta_r^{n-m} (m+1)^{-3/2}$. This is $O(\xi^n)$ for any fixed $\max(v, \eta_r) < \xi < 1$. The omitted series tail has the same type of bound. Applying the estimate to absolute coefficient differences before normalization proves the total-variation assertion. $\square$

The free energy, meaning the exponential growth rate of the weighted count, is therefore

$$f_r(y) := \lim_{n \rightarrow \infty} \frac{1}{n} \log Z_{n,r}(y) = \begin{cases} \log 4, & 0 < y \leq y_c, \\ \log(y/\rho_r), & y > y_c. \end{cases} \quad (21.3)$$

Its one-sided derivative with respect to $\log y$ jumps from zero to one. In that precise sense the transition is first order. The critical endpoint belongs to the late-failure phase in the law of K/n , despite the critical deficit's heavy tail.

For comparison, at $r = 2$ both $\Delta_2(xy)$ and $1 - xyC(x)$ vanish at $(x, y) = (1/4, 2)$. Directly from (16.2), with $z = \sqrt{1 - 4x}$,

$$F_2(x, 2) = \frac{1}{2z^3} - \frac{1}{z^2} + \frac{3}{2z} - 1.$$

Thus $Z_{n,2}(2) = (n+2)\binom{2n}{n} - 4^n$, the prior identity (9.2) of Part I. Its leading prefactor is $\sqrt{n}$, rather than the constant prefactor in (20.1). The extra vanishing factor explains why the $r = 2$ beta law cannot be extended unchanged to longer patterns.

[write] Added 5 October 2026, batch 98: the case $r = 2$. The displayed $F_2(x, 2)$ is Part I's (9.6), and the coefficient extraction is the one in the proof of Theorem 9.1. Although Theorem 21.1 is proved only for $r \geq 3$, its formulas hold at $r = 2$ by Part I's Theorem 10.2: with $v = 2/y$, the computation in the previous [write] note gives $\mathcal{P}_2(v) = \frac{1}{2} + \frac{C(\rho)^3}{8y\{1-C(\rho)/2\}} = \mathcal{A}(y)$ with $\rho = 1/(2y)$, and $p_{2,m}v^m/\mathcal{P}_2(v)$ is the law (10.4); also $y/\rho_2 = 2y$, so (21.3) at $r = 2$ is (10.6). (The amplitude identity was checked to 50 digits at $y = 3, 5, 7$ in this write.)

22 A displaced coexistence window

The critical limiting law describes $y = y_c$. It does not locate the finite-size region in which early and late failures both have substantial probability. We first prove a general endpoint-sum lemma, then apply the uniform estimate.

22.1 Competition between a summable tail and a geometric weight

Lemma 22.1 (Endpoint competition). *Let $p_m \geq 0$, $P = \sum_{m \geq 0} p_m \in (0, \infty)$, and suppose $p_m \sim am^{-\alpha}$, with $a > 0$ and $\alpha > 1$. For complex s , set*

$$w_n(s) = (\alpha - 1) \log n + \log \log n + s, \quad S_n(s) = \sum_{m=0}^n p_m \exp \left[-w_n(s) \frac{n-m}{n} \right].$$

Then, locally uniformly for $s \in \mathbb{C}$,

$$n^{\alpha-1} \log n S_n(s) \longrightarrow Pe^{-s} + \frac{a}{\alpha-1}. \quad (22.1)$$

The part with $m \leq n/2$ contributes Pe^{-s} ; the other part contributes $a/(\alpha-1)$.

Proof. All estimates below are uniform when s lies in a fixed compact set. Write $w = w_n(s)$ and $L = (\alpha-1) \log n$. For $m \leq n/2$, factor out e^{-w} . Up to $m \leq n/L^2$, $e^{wm/n} = 1 + o(1)$ uniformly, and the partial sum of p_m tends to P . On the remaining interval,

$$\sum_{n/L^2 < m \leq n/2} p_m |e^{wm/n}| \leq e^{\Re w/2} O((n/L^2)^{1-\alpha}) = o(1).$$

The last estimate follows because its power of n is $-(\alpha-1)/2 < 0$, whereas its other factors are powers of $\log n$. This proves that the late endpoint contributes $e^{-w}(P + o(1))$.

For $m > n/2$, put $k = n - m$. Uniformly on $0 \leq k \leq n/\sqrt{L}$, $p_{n-k} = an^{-\alpha}(1 + o(1))$. The remaining weighted sum is negligible: on $n/\sqrt{L} < k < n/2$, the coefficients are $O(n^{-\alpha})$, and the sum of exponential weights is at most $O(ne^{-c\sqrt{L}})$, for a fixed $c > 0$. For the principal range the geometric sum gives

$$\sum_{k \geq 0} e^{-wk/n} = \frac{1}{1 - e^{-w/n}} \sim \frac{n}{w}.$$

Absolute sums obey the same bounds, so the uniform coefficient error does not depend on cancellation when s is complex. Hence the early endpoint contributes $an^{1-\alpha}/w(1 + o(1))$. Now $e^{-w} = n^{1-\alpha}(\log n)^{-1}e^{-s}$, and $w/\log n \rightarrow \alpha-1$, proving (22.1). $\square$

For real s , the same proof gives two conditional laws for the probability weights defining S_n . Conditional on $m \leq n/2$, the variable m converges in total variation to p_m/P . Conditional on $m > n/2$, the variable $k = n - m$ is asymptotically geometric with success probability $1 - e^{-w/n}$, in total variation, and $(w/n)k \Rightarrow \text{Exp}(1)$. To see the variation assertions, use the same two cutoffs: on the principal range the relative weights are uniform approximations, while the absolute mass of the complements tends to zero. This also shows that the conclusions remain valid under an error whose total absolute mass is negligible relative to S_n .

22.2 The window and its two conditional limits

For $n \geq 3$, define

$$b_n = \frac{1}{2} \log n + \log \log n, \quad y_n(s) = y_c \exp \left[-\frac{b_n + s}{n} \right]. \quad (22.2)$$

Theorem 22.2 (The logarithmically displaced window). *For every fixed $r \geq 3$, locally uniformly for complex s ,*

$$\boxed{\sqrt{n} \log n 4^{-n} Z_{n,r}(y_n(s)) \longrightarrow 2a_r + P_r e^{-s}}. \quad (22.3)$$

For real s , under the probability measure with weight $y_n(s)^K$,

$$\frac{K_{n,r}^{(y_n(s))}}{n} \Rightarrow (1 - \omega_r(s))\delta_0 + \omega_r(s)\delta_1, \quad \omega_r(s) = \frac{P_r e^{-s}}{2a_r + P_r e^{-s}}. \quad (22.4)$$

Writing $\epsilon_n = (b_n + s)/n$, which is positive for all sufficiently large n for each fixed real s , the early phase satisfies

$$d_{\text{TV}} \left(\mathcal{L}(K_{n,r}^{(y_n(s))}) \mid K_{n,r}^{(y_n(s))} \leq n/2, \text{Geom}_0(e^{-\epsilon_n}) \right) \longrightarrow 0, \quad (22.5)$$

and hence $\epsilon_n K \Rightarrow \text{Exp}(1)$ under that conditioning. The late phase satisfies

$$d_{\text{TV}} \left(\mathcal{L}(n - K_{n,r}^{(y_n(s))}) \mid K_{n,r}^{(y_n(s))} > n/2, \mathcal{L}(M_r) \right) \longrightarrow 0. \quad (22.6)$$

Proof. By (19.7), uniformly for s in a fixed compact subset of $\mathbb{C}$,

$$4^{-n} Z_{n,r}(y_n(s)) = \sum_{m=0}^n p_{r,m} e^{-(b_n+s)(n-m)/n} + O_r(n^{-3/2}). \quad (22.7)$$

Indeed $\Re(b_n + s) > 0$ for all sufficiently large n , so the absolute exponential weight is at most one; the error is bounded by (20.5). Apply Lemma 22.1 with $\alpha = 3/2$, $a = a_r$, and $P = P_r$. The scaled error is $O((\log n)/n)$, which tends to zero. This proves the asserted complex-uniform limit, not only its real restriction.

For real s , the two endpoint contributions in the lemma are positive. Their normalized limits give the two weights in (22.4). The coefficient errors have total absolute mass $O(n^{-3/2})$, negligible compared with the mass of either endpoint, which has order $n^{-1/2}/\log n$. Thus the lemma's conditional variation estimates pass to the actual permutation distribution. The possible boundary term at $K = n/2$ has negligible mass, so its assignment to the early phase does not affect the conditional limits. In the early phase, $K/n \rightarrow 0$ because $n\epsilon_n \rightarrow \infty$; in the late phase, the deficit is tight. This proves all assertions. $\square$

The center can be normalized to eliminate the model-dependent constants. Put

$$s_r^* = \log \frac{P_r}{2a_r}, \quad \tilde{y}_n(s) = y_c \exp \left[-\frac{b_n + s_r^* + s}{n} \right]. \quad (22.8)$$

Then the late-phase weight is exactly the limiting logistic expression $1/(1 + e^s)$, and the limit in (22.3), divided by $2a_r$, is $1 + e^{-s}$.

The displacement is essential. If instead $y_n = y_c e^{-c/n}$ for a fixed real c , dominated convergence in the convolution gives $4^{-n} Z_{n,r}(y_n) \rightarrow P_r e^{-c}$, and the deficit still converges to M_r . Such a window sees only the late phase.

The reason for the extra $\log \log n$ is explicit. Late failures have mass about $P_r e^{-w}$, while early failures have mass about $a_r n^{-1/2}/w$. Equating them gives $w e^{-w} = (a_r/P_r) n^{-1/2}$. The large solution is expressible

using the negative real branch of Lambert W ; expanding that equation gives precisely $b_n + s_r^* + o(1)$. This is a way to understand the center, not an exact finite-size formula for the weight giving a half-and-half split.

[write] Added 5 October 2026, batch 98: relation to Part I's window. Part I's Theorem 10.4 is the $r = 2$ analogue: there the window $y = 2e^{\lambda/n}$ has order $1/n$ and is not displaced, and the limit is a continuous tilt (10.8) of the beta law, not a two-point mixture. Theorem 22.2 does not apply at $r = 2$ (Lemma 22.1 needs $P < \infty$; see the [write] note in Section 19). The Lambert- W reading of the centre in the previous paragraph is a heuristic of the manuscript; it is carried as an open item in Section 26 (question 3).

23 Complex zeros near the displaced window

The coefficients of $Z_{n,r}(y)$ are nonnegative, so it has no positive real zeros. The complex version of Theorem 22.2 nevertheless locates zeros approaching the positive critical point from off the real axis.

Theorem 23.1 (Local complex zero asymptotics). *Fix $r \geq 3$ and an integer j . For all sufficiently large n , there is a zero $z_{n,j}$ of $Z_{n,r}$ satisfying*

$$z_{n,j} = y_c \exp \left[-\frac{\frac{1}{2} \log n + \log \log n + \log(P_r/(2a_r)) + (2j+1)\pi i + o(1)}{n} \right]. \quad (23.1)$$

More precisely, in the s -coordinate of (22.2), there is exactly one zero, counted with multiplicity, in each sufficiently small fixed disk about $s_r^ + (2j+1)\pi i$, for all large n . On any compact region whose boundary contains no zero of $2a_r + P_r e^{-s}$, the total number of zeros eventually equals that of the limiting function.*

Proof. The entire functions $\varphi_n(s) = \sqrt{n} \log n 4^{-n} Z_{n,r}(y_n(s))$ converge locally uniformly to $\varphi(s) = 2a_r + P_r e^{-s}$. The zeros of φ are precisely the displayed s -values, and are simple because $\varphi'(s) = 2a_r$ there. On the boundary of a sufficiently small disk about one of them, $|\varphi|$ has positive minimum. For large n , $|\varphi_n - \varphi| < |\varphi|$ on the boundary, so Rouché's theorem gives one zero inside. Shrinking the disk proves convergence of its location. The exponential map defining $y_n(s)$ is locally one-to-one and has nonzero derivative, so multiplicity is preserved and the image is a zero of $Z_{n,r}$. The same boundary argument proves the compact-region assertion. $\square$

In particular, the pair corresponding to imaginary coordinates $\pm\pi$ has imaginary distance asymptotic to $\pi y_c/n$ from the real axis, while its real displacement from y_c is of order $(\log n)/n$. The theorem concerns these local sequences of zeros. It does not assert that they are globally the closest zeros, describe all zeros of $Z_{n,r}$, or settle the original $r = 2$ zero problem.

24 A worked example: first failure of 1432

For $r = 3$, the algebraic constants simplify to

$$\rho_3 = \frac{3 - \sqrt{5}}{2}, \quad \Lambda_3 = \frac{3 + \sqrt{5}}{2}, \quad y_c = 6 - 2\sqrt{5}, \quad (24.1)$$

$$P_3 = \frac{5 - \sqrt{5}}{2}, \quad a_3 = \frac{4}{\sqrt{\pi}}, \quad s_3^* = \log \frac{(5 - \sqrt{5})\sqrt{\pi}}{16}. \quad (24.2)$$

Numerically $y_c = 1.5278640450\dots$ and $s_3^* = -1.1835694676\dots$. The critical deficit generating function is

$$\mathbb{E} t^{M_3} = \frac{1}{5} + \frac{\sqrt{5} - 2}{20} \frac{tC(t/4)^4}{1 - \frac{3-\sqrt{5}}{2}C(t/4)}. \quad (24.3)$$

Thus exactly one fifth of the limiting critical mass has zero deficit:

$$\mathbb{P}(M_3 = 0) = \frac{1}{5}, \quad \mathbb{E}M_{n,3}^{(y_c)} \sim \frac{16}{\sqrt{\pi}(5 - \sqrt{5})} \sqrt{n} = 3.266011343 \dots \sqrt{n}. \quad (24.4)$$

The equality $M = 0$ means that the entire permutation avoids 1432. The remaining mass has a heavy tail, even though $K/n \rightarrow 1$.

n	Row $T_3(n, 1), \dots, T_3(n, n)$	Row sum
3	0, 0, 5	5
4	0, 0, 1, 13	14
5	0, 0, 4, 4, 34	42
6	0, 0, 14, 17, 12, 89	132
7	0, 0, 48, 62, 53, 33, 233	429
8	0, 0, 165, 219, 199, 149, 88, 610	1430

Table 8: Exact rows from the positive formula, independently checked by direct permutation inspection and an insertion-state recurrence.

n	$\mathbb{E}_{y_c} M / \sqrt{n}$	$\mathbb{P}_{y_n(0)}(K > n/2)$
100	1.8319121364	0.4402574792
400	2.4626378672	0.3918548955
1200	2.7801973301	0.3667208049
4000	2.9925079412	0.3427198278
Limit	3.2660113430	0.2344110035

Table 9: Convergence at the critical point and at $s = 0$ in the uncentered window (22.2). The limiting window weight at $s = 0$ is $P_3/(P_3 + 2a_3)$, not one half.

For additional perspective, numerical root finding with exact polynomial coefficients gives the values in Table 10. They approach the root coordinate predicted by Theorem 23.1, but slowly. Small numerical residuals validate the displayed approximate roots; they do not certify their global ordering among all polynomial zeros.

n	$\Re_s$	$\Im_s$
100	-0.6966203849	5.4649985806
400	-0.8067427947	4.8600345465
1200	-0.8602682888	4.5128712955
4000	-0.9150779479	4.2367353381
Predicted limit	-1.1835694676	3.1415926536

Table 10: Numerical zeros near the first upper-half-plane branch in the s -coordinate. The computations use 55 decimal digits; they are diagnostic, not interval-certified root enclosures.

[write] Added 5 October 2026, batch 98: data sources. Tables 9 and 10 agree digit for digit with the shipped files `critical_moments.csv`, `coexistence_probabilities.csv` and `complex_zero_diagnostics.csv`, and the rows of Table 8 are in `exact_small_tables.csv` (all four in `data/`, with the prefix `02-longer-patterns-`). The intake's independent check reproduced ρ_3 , y_c , P_3 , a_3 , s_3^* , $2a_3/P_3$, $P_3/(P_3 + 2a_3)$ and $\mathbb{P}(M_3 = 0) = \frac{1}{5}$ to 40 digits.

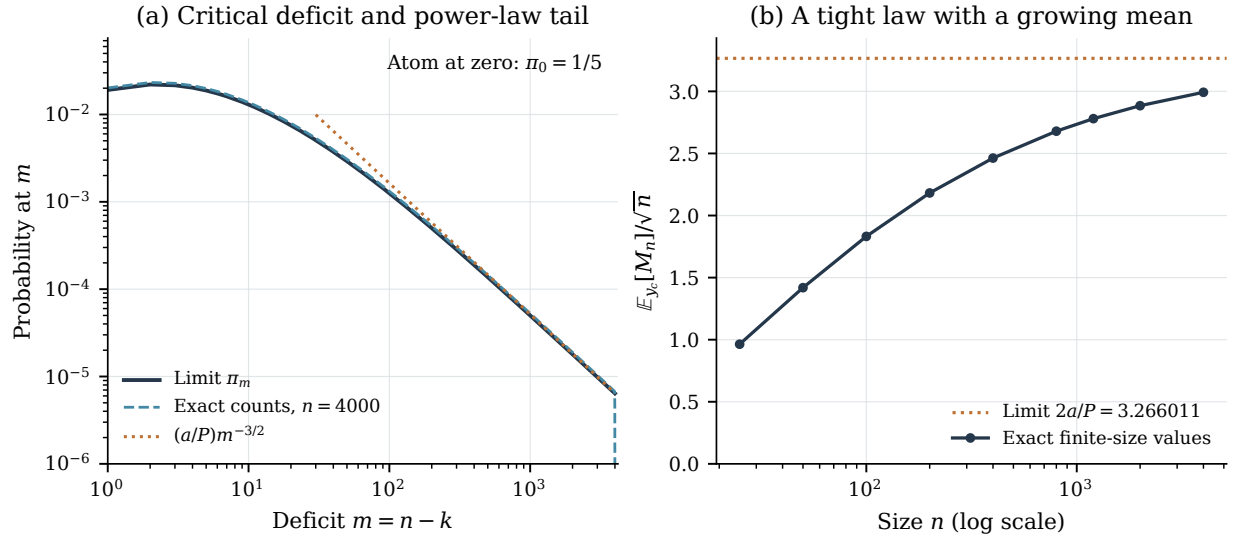


Figure 4: Critical behavior for $r = 3$. The left panel compares the finite-size deficit masses at $n = 4000$, the exact limiting generating function (24.3), and its power-law tail. The right panel shows the slowly converging normalized mean. Counts are computed with integer arithmetic before numerical normalization. The curves are diagnostics of the proved limits, not finite-input error certificates.

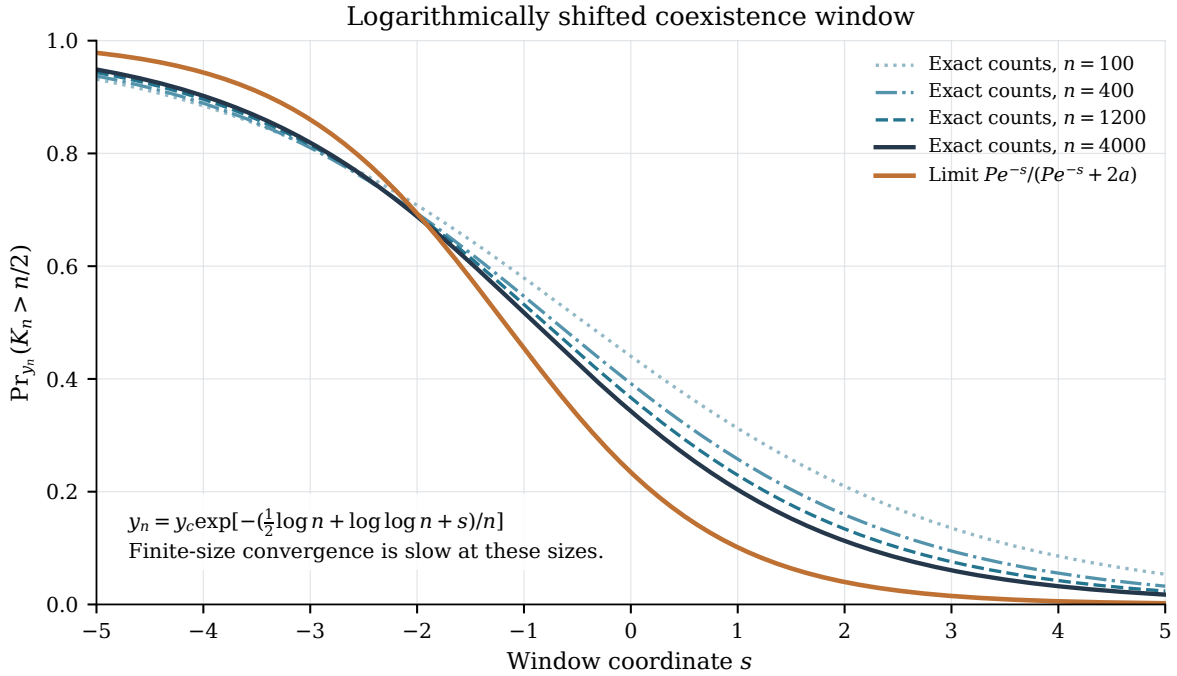


Figure 5: The late-failure probability for $r = 3$ in the displaced window (22.2). Finite-size values use exact integer coefficients; the limit is $\omega_3(s)$. Even at $n = 4000$, the finite curves remain visibly separated from the limit. The theorem asserts convergence and does not provide a small finite-size error at these sizes.

25 Verification, reproducibility, and formalization

The proofs above are independent of the computations. Finite enumeration can detect errors in the model, indices, or formulas, while the uniform estimate is what establishes the asymptotic conclusions. The archive contains two complementary programs using only the Python standard library for their core verification.

Independent check	Executed scope
Literal subsequence tests on permutations	Every permutation of sizes 1–9; $r = 2, \dots, 6$.
First-failure and suffix-state insertion DP	$n \leq 40, r = 2, \dots, 6$.
Safe-state tails, support, diagonal, Catalan sums	$n \leq 120, r = 2, \dots, 6$.
Direct series substitution in the bivariate GF	Five integer values of $y, n \leq 40, r = 2, \dots, 6$.
Rational normalization, mean, variance	$r = 2, \dots, 60$, with independent spectral-sum checks.
Integer recurrence for the figures	Selected $r = 3$ rows through $n = 4000$; 166 entries independently checked by the finite binomial sum.

Table 11: Verification actually executed. Detailed counts and outcomes are preserved in the JSON records.

The literal enumeration inspected 409,113 permutations, including all 6,917 permutations avoiding 123 in those sizes. The figures’ program performed 7,990,003 integer recurrence updates and verified the Catalan sums of all nine rows it computed. Numerical normalization and plotting then use floating-point arithmetic; the optional complex roots use `mpmath`. The additional limit-law audit includes large-index critical-moment comparisons and the exact arithmetic underlying the Wasserstein bound. The saved output distinguishes these comparisons from exact rational identities.

25.1 Reproduction

From the extracted article directory, run

```
python3 code/verify_first_failure.py
python3 code/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The first command uses only the standard library and also runs `audit_limit_laws.py`. The figure command needs the dependencies listed in `requirements.txt`. The included figure PDFs and data can be used directly, so rebuilding the article alone does not require Python. Programs write fresh outputs to `data/` and `figures/`; the delivery README describes the output paths. No program needs a network connection.

[write] Added 5 October 2026, batch 98: the shipped layout. The commands above are for the delivered archive. In this report the manuscript’s files carry the prefix `02-longer-patterns-` in `code/`, `data/` and `figures/`, with `requirements.txt` and `provenance.json` in `data/`; the delivery README, article and PDF are not shipped. The two verification programs import each other by their delivered names, and every program writes unprefixed files under `data/` and `figures/`: run in place, `verify_first_failure.py` would overwrite Part I’s `data/verification_summary.json`. Run them only on a copy with the delivered names restored; the README gives the commands. Rerun so on 5 October 2026 (Python 3.14.4), the verifier passed in 39 s with its CSV byte-identical, its summary identical up to the elapsed time, and the limit-law audit identical up to last-place floating-point digits; the figure program passed in 3 min 16 s with the zero diagnostics byte-identical and the other tables equal to relative $9 \cdot 10^{-16}$.

25.2 A concrete route to proof-assistant formalization

The first formalization target should be the combinatorial meaning of the coefficients. Define pattern occurrence using order-preserving maps of finite position sets, and define restriction to the largest k values. Prove that inserting a new minimum and deleting it are inverse operations after standardization. Then establish the active-site lemma and show that a first τ_r occurrence is created exactly at sites $j \geq r$. These steps connect the count to permutations; a formal identity between rational generating functions alone would not make that connection.

The next layer is algebraic: safe-state recurrences, the tail $u^j/\Delta_r(u)$, the Catalan descendant series, and the coefficient formula. It can be checked with formal power series and finite sums. The Chebyshev factorization and the geometric distribution give further isolated targets, including exact moments and the coupling inequality.

The analytic layer should state (19.7) with explicit quantifiers over both indices and over constants depending on r . From that interface, the critical laws and the endpoint lemma use ordinary estimates for positive series. The complex-zero result adds local uniform convergence and Rouché’s theorem. Nothing in this article is claimed to have been formalized in Lean or Rocq, and the repository’s other formal developments do not change that status.

26 Further questions and research

The following questions are specific continuations of the results proved here. They are proposed research directions; except where a source is named, they are not represented as established open problems from the literature.

[write] **Added 5 October 2026, batch 98: how this list is kept.** Questions 1–10 are manuscript 07’s section 12, printed in full. By the repository’s rule for unproved claims, questions 11 and 12 add the two assertions of the manuscript that this Part does not prove (each with its source, its sketch and what is missing). Where a question coincides with one of Part I’s (Section 13), both are kept and each carries a note. No claim of the manuscript was found to be false.

1. **A joint large-permutation, long-pattern theorem.** Determine conditions on $r = r(n)$ under which $K_{n,r}^{(1)}/r^2 \Rightarrow \mathcal{W}$. The natural starting range is $r^2 = o(n)$, but the current fixed- r error constants are not uniform there. When r is of order $\sqrt{n}$, the finite ceiling $K \leq n$ interacts with the limit law and should create a new crossover. No particular joint limiting formula is asserted here.
2. **Uniformity as the critical weight approaches one.** Since $y_c = \sec^2(\pi/(r+2)) = 1 + \pi^2/(r+2)^2 + O(r^{-4})$, the weighted transition moves toward the unweighted model as the pattern grows. Develop a theory in which this displacement competes with the $(\log n)/n$ displacement of Theorem 22.2. This requires keeping the spectral gap and the tail constants uniform in r .
3. **Higher-order coexistence asymptotics.** Expand the endpoint convolution beyond its limit, including terms in $(\log \log n)/\log n$, $1/\log n$, and $n^{-1/2}$. The sizeable discrepancies in Table 9 make this practically useful. A worthwhile endpoint is a rigorously controlled approximation to the actual finite-size median transition weight.

[write] **Added 5 October 2026, batch 98: the centre, what is proved and what is not.** The manuscript explains the centre of the window by the equation $we^{-w} = (a_r/P_r)n^{-1/2}$, whose large (Lambert- W_{-1}) solution expands to $b_n + s_r^* + o(1)$ (Section 22), and calls this “not an exact finite-size formula”. Its leading order is in fact a consequence of Theorem 22.2. For fixed n , the tilted laws of K form an exponential family in $\log y$, so $\mathbb{P}_y(K > n/2)$ is continuous and strictly increasing in y , from 0 to 1; let y_n^{med} be the weight where it equals $\frac{1}{2}$, and write it as $\tilde{y}_n(\sigma_n)$ in the

centred coordinate (22.8). The proof of Theorem 22.2 shows that $\mathbb{P}_{\tilde{y}_n(s)}(K > n/2) \rightarrow 1/(1 + e^s)$ for each real s . For $\varepsilon > 0$ the limits at $s = \mp\varepsilon$ lie on either side of $\frac{1}{2}$, so by monotonicity $|\sigma_n| < \varepsilon$ for all large n : $\sigma_n \rightarrow 0$, that is, $y_n^{\text{med}} = y_c \exp[-(b_n + s_r^* + o(1))/n]$. What remains open is the rate of $\sigma_n \rightarrow 0$, which is this question; at $r = 3$ the finite-size late probability at $s = 0$ of the uncentred window is still 0.343 at $n = 4000$ against the limit 0.234 (Table 9).

4. Effective constants in the uniform estimate. The proof permits a concrete contour radius R and explicit majorants. Optimize these choices to produce usable numerical values of Θ_r and η_r , then propagate them through the critical and coexistence estimates. This would turn the present asymptotic laws into certified finite-input error bounds.

5. The original $r = 2$ complex-zero problem. The source report asks about zeros near $y = 2$. The present zero theorem applies to $r \geq 3$; $r = 2$ has a different limiting function because of the additional vanishing factor. Prove its complex-uniform scaling limit and analyze the resulting entire function's zeros, preserving that distinction.

[write] Added 5 October 2026, batch 98. This is Part I's question 4 (Section 13), which stays open; Part I proposes $M(\lambda)$ of (10.7) as the governing function. At $r \geq 3$ the corresponding function is $2a_r + P_r e^{-s}$ (Theorem 23.1).

6. Global zeros for the longer-pattern family. Determine whether the local pair in Theorem 23.1 is eventually closest to the positive real axis in an appropriate region. Describe the other accumulation curves, and prove a zero-counting measure limit. The local Rouché argument gives no such global information.

[write] Added 5 October 2026, batch 98. The 55-digit roots of Table 10 (scaled residuals at most 10^{-55} in `complex_zero_diagnostics.csv`) are not interval enclosures, and neither their position among all zeros of $Z_{n,3}$ nor the slow approach to the predicted coordinate ($\Im s = 4.24$ at $n = 4000$ against π) is proved; both belong to this question.

7. Joint statistics at the failure event. Mark the suffix length immediately before failure, the failing site, and the final suffix length. The proof of Theorem 16.1 retains the site variable j , so a multivariate refinement is accessible. Determine its dependence on the heavy-tailed deficit and on the early-versus-late phase indicator.

[write] Added 5 October 2026, batch 98. This generalizes Part I's question 2 (joint first-failure and suffix statistics at $r = 2$) to every r and adds the phase indicator; neither is answered.

8. A direct path interpretation of first failure. Chow–West and Krattenthaler identify the safe counts with bounded-height Dyck paths. Find a bijection carrying K_r to a natural path stopping statistic, not merely preserving the total number of safe objects. It may explain the geometric spectrum and the law $\mathcal{W}$ at the level of individual combinatorial objects.

[write] Added 5 October 2026, batch 98. Part I's question 7 asks the same at $r = 2$.

9. Other forbidden-pattern families. Find classes whose safe transfer matrix has a dominant eigenvalue strictly below the unrestricted Catalan growth constant, and whose continuation law has a summable power tail. Establish hypotheses under which a uniform analogue of (19.7) holds. Then Lemma 22.1 predicts the displacement $(\alpha - 1) \log n + \log \log n$ in the exponent for a tail $m^{-\alpha}$. The key difficulty is deriving the uniform kernel, not applying the endpoint lemma afterward.

10. Kernel-checked enumeration and certified sampling. Formalize the positive coefficient formula and use it for exact sampling conditioned on $K = k$. The independent geometric representation already samples the infinite-size subcritical law. Compare exact finite-size sampling with that approximation, and make the error bounds executable.

[write] **Added 5 October 2026, batch 98.** Part I’s question 8 asks for the $r = 2$ exact layer in Lean; nothing in either Part is formalized.

11. **The predicted displacement in other families** ([write], from question 9 and Lemma 22.1). The manuscript’s “predicts” in question 9 is conditional: the displacement $(\alpha - 1) \log n + \log \log n$ follows from Lemma 22.1 only for a family that has a uniform estimate of the form (19.7) with a tail $p_m \sim am^{-\alpha}$ of finite total mass. No family other than τ_r , $r \geq 3$, is shown here to have one, and the [write] note in Section 19 shows that τ_2 does not. What is missing is the uniform kernel for any second family.
12. **Priority and the cited theorem numbers** ([write], from the manuscript’s Section 27). The manuscript reports a targeted literature search that found no treatment of the longer-pattern first-failure refinement; it does not claim worldwide priority, and this write did not extend the search. The attributions of the safe counts to Chow–West, Theorem 3.1, and of the Dyck-path formulation and occurrence marking to Krattenthaler, Theorems 9 and 8, are the manuscript’s; this write did not recheck the theorem numbers against the papers. The reverse-complement step that matches τ_r with Chow–West’s pattern $r(r - 1) \cdots 1(r + 1)$ was rechecked.

27 Source audit and statement boundaries

The repository was read at 0f93381e8932d3aee3d5c2abfb5c98d128d49b9c. The full host article, its README, the research-report catalogue, and the checked-out enumerative and generating-function research directories were searched for this statistic. The inspected host still lists the general family in its first proposed research question; no later treatment was found in that scoped search. This is a snapshot-specific report, not a claim about every branch or uninspected file.

[write] **Added 5 October 2026, batch 98.** Rechecked at the placement commit: outside this report, the repository mentions A273821 only in the collection’s README and manifest, the intake README, a review catalogue and a .gitattributes file (listings and paths, not treatments); the pattern $1(r + 1)r \cdots 2$ occurs only here, and no Lean or Rocq file mentions A273821. The host’s question is now answered by this Part (the dated note at Part I’s question 1).

The direct OEIS retrieval identified the original $r = 2$ triangle and its displayed prefix. The safe enumeration was cross-checked against the primary Chow–West and Krattenthaler papers. The literature search also checked the first-ascent predecessor and standard singularity analysis. It found no independent treatment of the exact longer-pattern first-failure refinement, but does not establish worldwide priority.

The distinctions relevant to reuse are:

- Theorem 16.1 is exact for every integer $r \geq 2$. The $r = 2$ specialization is prior within ProveIt.
- Theorems 19.2–23.1 fix $r \geq 3$. Their constants have no claimed uniformity as $r \rightarrow \infty$.
- The growing-pattern theorem applies to K_r^∞ , the already-taken $n \rightarrow \infty$ limit. It does not commute the limits.
- The complex-zero theorem is local in the scaled variable, for each fixed zero index. It does not assert a global ordering of roots.
- Numerical values and small root residuals are not interval certificates. The proofs of limit theorems are the analytic arguments, not those computations.

28 A short exact-moment calculation

Formula (18.4) expands as the polynomial identity

$$\Delta_r(u) = 2^{-r-1} \sum_{j \geq 0} \binom{r+2}{2j+1} (1-4u)^j, \quad \Delta_r^{(m)}(1/4) = 2^{-r-1} (-4)^m m! \binom{r+2}{2m+1}. \quad (28.1)$$

Terms with out-of-range binomial coefficients are zero. It follows that

$$\frac{\Delta_r'(1/4)}{\Delta_r(1/4)} = -\frac{2r(r+1)}{3}, \quad \frac{\Delta_r''(1/4)}{\Delta_r(1/4)} = \frac{4}{15} r(r+1)(r-1)(r-2).$$

These two identities determine the mean and variance of $\sum_j G_j$ by differentiating $\Delta_r(1/4)/\Delta_r(t/4)$ at $t = 1$. Adding $r + V_r$, where

$$\mathbb{E}V_r = 1 + \frac{2}{r+2}, \quad \text{Var } V_r = 2 + \frac{6}{r+2} - \frac{4}{(r+2)^2},$$

gives (17.8)–(17.9). This derivation also justifies the finite trigonometric sum used in the Wasserstein proof without importing a separate summation theorem.

A Additional exact identities and coefficients

A.1 The OEIS triangle recurrence

With the boundary conventions stated in Section 1, the proved generating function also verifies the recurrence displayed in the OEIS entry:

$$\begin{aligned} T(n, k) = & 3T(n-1, k-1) - 2T(n-2, k-2) \\ & + T(n, k+1) - 2T(n-1, k) + \mathbf{1}_{k=2} C_{n-2}, \quad n \geq 3, 2 \leq k \leq n. \end{aligned} \quad (\text{A.1})$$

Together with $T(1, 1) = 1$, $T(2, 2) = 2$, and the zero boundaries, it computes rows from right to left. It is a consequence of the counting proof, not an assumption used to derive it. One verification is to insert the binomial expression and use Pascal's identity; another is to multiply (1.4) by the recurrence kernel and compare coefficients.

A.2 A third fixed-column correction

The coefficient generator (5.9) gives

$$c_3(k) = -\frac{2k^7 - 38k^6 + 130k^5 + 270k^4 - 3657k^3 + 26093k^2 - 34725k + 13860}{3072(k+4)}. \quad (\text{A.2})$$

This coefficient is independently regenerated in the symbolic check, rather than merely inserted as a numerical fit. For higher orders, the power sums in (5.2) and the exponential recurrence (5.5) are preferable to increasingly large printed polynomials.

A.3 The second critical raw moment

A second application of $y\partial_y$ gives

$$(y\partial_y)^2 F|_{y=2} = \frac{1}{2z} - \frac{3}{2z^2} + \frac{2}{z^3} - \frac{3}{2z^4} + \frac{1}{2z^5} - \frac{1}{z^6} + \frac{1}{z^7}. \quad (\text{A.3})$$

Consequently,

$$\sum_{k=1}^n k^2 2^k T(n, k) = \frac{2(4n^3 + 23n^2 + 63n + 30)}{15} \binom{2n}{n} - \frac{(n+2)(n+4)}{2} 4^n. \quad (\text{A.4})$$

Together with (9.2) and (9.3), this provides an exact finite- n variance and another independent check on the limiting variance $4/45$ after scaling by n^2 .

B Source and claim ledger

Item	Status and use
A273821 definition and conjecture	Primary OEIS entry retrieved October 1, 2026. The entry labels its generating function conjectural and suggests fixed-column growth 4. The present report proves both claims from the permutation definition.
A000245 and A071718	Known companion columns with the initial exceptions specified in (4.8). Their existing leading asymptotics are acknowledged, not claimed as new.
Catalan insertion machinery	Proved directly in Sections 2–3. General Catalan and singularity-analysis methods are established background.
Stirling and transfer estimates	Standard analytic inputs, cited to DLMF and <i>Analytic Combinatorics</i> ; their hypotheses are checked in the uses above. Coefficients and model-specific deductions are derived here.
Exact formulas and limit laws	Conventional proofs supplied in this report; independent computational checks supplied in the archive. No Lean certificate is claimed.
ProveIt connection	The inspected transseries project supplies methodological context and a possible formalization destination, not an assumed proof of any theorem in this report. Repository tree inspected at commit <code>291f6e6bb4477e0cb2131f1e041b12a9abe363fd</code> .
Novelty search	Targeted identifier/statistic searches and repository searches found no independent proof of the posted conjecture. This is not an exhaustive bibliographic certification.
Proposed extensions	Section 13 distinguishes further research from theorems actually proved. The OEIS amendment file is a draft, not a submitted edit.

References

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- [5] NIST Digital Library of Mathematical Functions, §5.11: *Asymptotic Expansions of the Gamma Function*, including the logarithmic Stirling expansion and remainder estimates. Retrieved October 1, 2026. <https://dlmf.nist.gov/5.11>.
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- [7] ProveIt, *Combinatorial Transseries and Their Inverses*, repository volume `Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/Combinatorial_Transseries_Inverses.tex`, theorems labelled `t2:thm:gamma` and `t2:thm:balanced-inverse`. Not formally verified. Added in writing (batch 74).
- [8] ProveIt, *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and inversion*, repository volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`, theorem labelled `p0:thm:staircase`, which that volume records as partly formalized in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`. Added in writing (batch 74).
- [9] Vladimir Reshetnikov, *ProveIt*, transseries project README and repository tree inspected October 1, 2026. <https://github.com/VladimirReshetnikov/ProveIt>. Project path: `Analysis/Transseries/README.md`. The README records the independent transseries library, canonical inversion volume, and combinatorial companion; these are contextual rather than logical dependencies of the proofs here.
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